EurekaZero

Engineered Intelligence for Algorithm Invention by Self-Play

JopeDime Team

Draft — October 2026

Abstract

[TBD: Supply the problem statement, proposed method, supported task scope, contributions and evaluation summary from Jepson's source.]

[TBD: State the headline first-proposal result with its uncertainty once the independent evaluation is complete.]

1 Can Machines Invent?

We specify how to test whether feedback from execution improves the first algorithm a system proposes for a new problem. AlphaGo Zero supplies a precedent within a fixed game: it went beyond the best human play without seeing a human game, learning from self-play under the rules of Go [101].

The skeptics' challenge. Language models retrieve existing solutions rather than find new ones, Yann LeCun argued in March 2025: "in their purest form, they are incapable of inventing new things" [58]. Thomas Wolf warned that current trends will give us "a country of yes-men on servers" [117].

Dwarkesh Patel had asked Dario Amodei in August 2023 what to make of models with "basically the entire corpus of human knowledge memorized" that had not made "a single new connection that has led to a discovery" [80]. In January 2025, Demis Hassabis posed the hardest version as a benchmark for general intelligence: today's systems "can play a game of Go at a world champion level. But could a system invent Go?" [41]. These challenges concern both proposing solutions and choosing new questions; our test concerns only algorithm proposals under human-defined tasks and metrics (Section 8).

Predicting historical text does not directly reward improvement over an incumbent. That motivates a different training signal; it does not establish a ceiling on what a pretrained model can discover (Section 2).

The human challenge. Research effort rose while productivity declined in the industries, products and firms examined in the 2020 study by Bloom et al. [8]. Their Moore's law example required more than 18 times as many researchers as in the early 1970s to sustain chip-density growth. Jones [46] found later first inventions and greater specialization in the inventor data he studied. Park et al. [79] reported declining disruption on citation-based measures of papers and patents. These observations motivate our question: can execution feedback help a proposer depart usefully from familiar ideas? Section 2.4 develops the role of the scientific record.

Key idea

Can judgment bought during training improve the first proposal at deployment?

From search to learning. Automated research systems already couple proposals to tests [37, 49, 65, 75, 104, 105]. Sections 2.5 and 3 compare how they generate, judge and retain useful work.

The learning test asks whether evaluation feedback improves the first proposal on a new problem.

The Zero move. AlphaGo learned from human expert games and searched heavily at every move [100]. AlphaGo Zero learned only from games against itself, and its network alone, with no search, rated 3,055 Elo. That is close to the 3,144 of AlphaGo Fan, the version that beat the European champion Fan Hui [100, 101].

This motivates an analogous goal for invention: learn from repeated program evaluations during training, then deploy a writer with a single-candidate budget, search $= 1$ (Box 1). The budget is a test condition, not a promise that every first proposal wins. Figure 1 separates the training feedback path from the frozen writer and fitting procedure at deployment.

Engineered intelligence. The reference design assigns each role in Section 2 an inspectable component: a declared task and metric, a grammar-constrained writer, a referee and retained state (Table 1). Program-incumbent promotion and proposed writer admission are distinct rules in that design. We call the design *engineered intelligence*: its aim, proposer, referee and retained state are specified for audit. Pretrained components still supply prior knowledge (Box 1); execution feedback supplies the additional teacher.

Contributions. We formulate algorithm invention as a game and give a tabular-regression reference whose moves are typed programs of learning primitives.

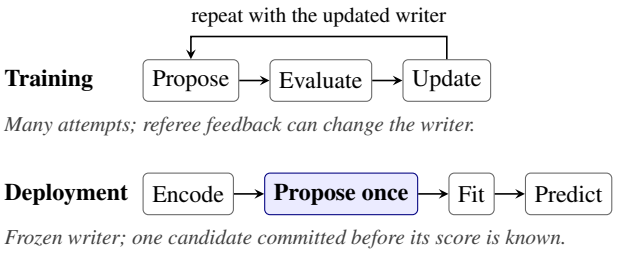


Figure 1: **Training feedback and the single-candidate contract.** Reference schematic: the proposed procedure updates the writer during training and fits the emitted estimator during deployment. The adapter-only record in Section 6.8 is a separate loop; first-proposal improvement remains the empirical question in Section 7.

Box 1: What "search = 1" means

- **One inference, one candidate.** For a new problem, the program writer emits one complete program ($B = 1$), with no retries, repairs, best-of-N, reranking or weight updates.
- **Everything is counted,** including reasoning tokens and problem encoding. One inference is one generation, not one forward pass.
- **Declare the fitting procedure.** Fitting the emitted algorithm to training rows is allowed. Any internal model or hyperparameter selection must be disclosed and counted in its compute budget. One outer proposal does not establish the absence of internal search. Training search is priced separately in Section 5.

A single candidate, with prior knowledge: human-designed program primitives and pretrained encoding components supply prior knowledge. The Zero analogy concerns the additional training feedback from the referee.

1. **The anatomy of innovation:** four pillars (aim, variation, judgment and retention), with an idealized relation between proposal quality and the number of attempts (Section 2).
2. **The innovation game:** a board, move, referee and opponent, with a reward ordering and two reference admission rules; each task type needs a qualified encoding and referee (Section 4).
3. **The economics of algorithmic self-play:** a cost model for training and for the cost per verified improvement (Section 5).
4. **EurekaZero:** a reference learner with a frozen problem encoder, a grammar-constrained program writer and a separate problem–program map, plus an expert-iteration procedure and safeguards against reward hacking (Section 6).
5. **An evaluation protocol for the single-proposal claim:** paired comparisons with frozen algorithms whose receipts record the model that designed them, performance curves against agents allowed repeated attempts, and a test of the map's ranking against a program-only control (Section 7).

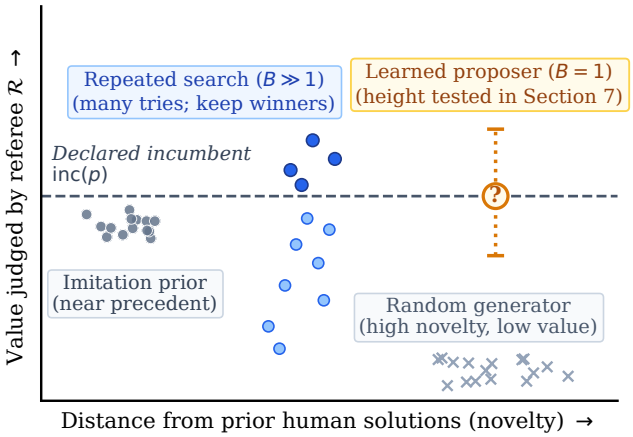


Figure 2: **Novelty is free; judgment is the inventor.** Schematic. Horizontal: distance from what people have already done. Vertical: value as judged by the referee. A random generator travels far and finds little. Imitation, repeated search and a learned proposer illustrate different ways to select candidates; their positions are conceptual, not measured limits. The value of EurekaZero's single proposal remains an empirical question for Section 7.

Evidence scope. The adapter records in Section 6.8 establish neither neural writer updates nor a training gain in the separate ranking map. Independent first-proposal comparisons remain pending; Sections 6 and 7 specify the reference design and its tests.

We first separate innovation's roles (Section 2), then trace the AI methods that connect them (Section 3). The game defines an attempt (Section 4); the cost model prices it (Section 5); and the reference learner connects attempts across tasks (Section 6). The tests in Section 7 will bound the claims in Section 8.

2 The Anatomy of Innovation

The learning question in Section 1 requires separating four roles: aim, variation, judgment and retention. They connect what a system proposes, how it is evaluated and what the next attempt retains. We call judgment learned before a try *taste*.

2.1 Novelty is free

Novelty is the cheapest part of invention. Borges imagined a library of every possible book of 410 pages [11]. Every algorithm short enough to fit in a volume is already on its shelves. Nothing in the library tells a reader which volume is true. A random string generator is the same library with a faster librarian: almost every string it emits is new, and almost none is worth running (Figure 2, lower right).

Language-model studies test this distinction. In a blinded study with 104 NLP researchers, ideas from a language model were rated more original than experts' ideas ($p < 0.05$) [98]. Once 43 experts had spent over 100 hours each carrying out ideas from both pools, the generated ideas lost significantly more ground on every metric [99].

Table 1: **A four-part reading of traditions of discovery.** A dash marks a pillar outside a tradition's scope: Wallas analyzed a single act of thought. For Darwin and Campbell, the aim is a criterion set by the environment. Engineering is the design–build–test–learn cycle [74]; AlphaGo Zero follows Silver et al. [101]; EurekaZero is the reference design (Sections 4 and 6).

Tradition	Aim	Variation	Judgment	Retention
Darwin [25]	Fitness to the environment	Variation	Natural selection	Inheritance
Wallas [115]	Preparation	Incubation, illumination	Verification	—
Campbell [15]	A selection criterion	Blind variation	Selection	Retention
Popper [87]	The problem	Conjecture	Refutation	The surviving theory
The scientific method	Question	Hypothesis	Experiment	Publication
Engineering	Design target	Design and build	Test	Learn
AlphaGo Zero	Win the game	Moves chosen with search	Game outcome	Network weights
EurekaZero	Problem encoding and metric	Program writer	Referee on hidden folds	Incumbent pool; proposed writer updates

Boden had predicted the asymmetry in 1998: machines, she wrote, would find generating new ideas easier than automating their evaluation [10]. These studies motivate a distinction between the novelty of a proposal and its value after execution; they do not establish that judgment is the sole bottleneck.

2.2 The four pillars

Accounts of discovery from biology, psychology, philosophy and engineering provide a useful common vocabulary (Table 1). The four-pillar organization is our synthesis of these traditions, rather than a claim that their theories are identical.

Campbell argued that every genuine gain in knowledge needs a mechanism for introducing variation, a consistent selection process, and a mechanism for preserving and reproducing what is selected [15, p. 381]. The aim is what keeps the selection consistent, which makes it the fourth pillar.

Aim fixes the problem and what counts as better. Hamming asked colleagues, "What are the important problems of your field?" [40]. Pólya's first step is to understand the problem [85], and Popper argued that observation is always selective, so it presupposes a problem [87].

Variation proposes candidates. Peirce called abduction "the only logical operation which introduces any new idea" [81, CP 5.171]. Campbell modeled creative thought as blind variation [15, 103]. Koestler located creative acts in bisociation, joining two unrelated frames of thought [52]. Boden named three routes to a new idea: combining familiar ideas, exploring a conceptual space and transforming it [10].

Judgment tests a candidate against something outside the mind that proposed it: the attempted refutation [86], the experiment, peer review, and the referee of the common task framework, which scores every entry on data the competitors never see [29].

Retention preserves what survives: inheritance for Darwin [25], selective retention for Campbell [15], the textbook for a science and the incumbent for a game. It can store either an output or a way to propose better outputs. Archiving a winning program does the first; using evaluation feedback to update the proposer attempts the second. This distinction connects the four pillars to the learning question in Section 1.

2.3 Taste is judgment moved upstream

Skill in invention is judgment that arrives before the try. For Poincaré, creating meant not making the useless combinations: "Invention is discernment, choice." The sterile combinations, he added, do not even present themselves to the inventor [83, pp. 324–325].

Chess masters show the same economy: in de Groot's think-aloud studies they considered about as many moves as weaker players but found the right ones [26]. Chase and Simon studied chunks through perception and recall tasks [18]. Much of that knowledge is tacit; experts know more than they can say [84]. AlphaGo's policy network learned to select moves from human expert games, then refined that judgment through self-play [100].

Taste has a price we can state. Suppose each try beats the incumbent with a fixed, nonzero probability q, independently of the others. The number of tries up to and including the first win is then geometric, with mean

$$\mathbb{E}[N_{\text{tries}}] = \frac{1}{q}, \tag{1}$$

where N_{tries} counts all candidate attempts, including incomplete or invalid outputs and failed executions. Under this fixed-probability, independent-try model, better proposal quality shortens the expected search.[1]

The equation prices proposal quality, whether training, structural constraints or revealed outcomes provide it. Comparing random generators, pretrained models and agent systems therefore requires measured proposal quality. In particular, search $= 1$ fixes the number of candidates a system may submit; it does not require the probability of a win to approach certainty. A single-candidate system can fail, and every failure must remain in its reported results. Section 5 uses the same

[1] Adaptive search and correlated proposals need not follow Equation (1); independence alone does not say whether the approximation overstates or understates their cost. Empirical attempt curves are required for those systems.

idealization to relate evaluation cost to the cost of a verified improvement.

Key idea

Novelty is free. Judgment is the inventor.

2.4 Learning from a referee

People learn much of their taste from the record of their field. Normal science, in Kuhn's account, solves puzzles inside a reigning paradigm [55]. The record also holds people in place: when a star life scientist dies prematurely, articles by outsiders to the star's collaborators flow into the field, draw on different literature and are disproportionately likely to be highly cited [2].

A model trained to predict text receives a signal about what was written, rather than a direct score for improving an algorithm. That does not prevent novel generalization. It motivates supplementing the historical record with feedback from execution, so a useful departure from precedent can be rewarded.

A referee offers such an additional teacher. AlphaGo Zero learned with no human data, from the rules and the outcomes of its own games, and found strategies outside traditional Go knowledge [101]. Its predecessor had already made such a move. As David Silver told Wired in March 2016, AlphaGo put the chance that a human would play its Move 37 against Lee Sedol at one in ten thousand, and it won that game [69].

The knowledge then flowed back to people. In more than 5.8 million moves by professionals from 1950 to 2021, previously unseen moves grew more frequent once superhuman programs arrived, and came with better decisions [97]. These results establish the possibility in Go. Whether an analogous signal improves algorithm invention must be measured in the innovation game.

2.5 Where the pillars live

Machines do not remove the pillars; they move them (Table 2). The most ambitious swarm design fits the same anatomy. In an October 2026 essay drawn from his book, Socher rests the Eureka Machine on "a living map of all human knowledge", "a unified model of physical reality", "high-fidelity digital twins" and "fully autonomous physical laboratories" [104, 105]. On top of them, millions of AI agents critique each other's ideas. In our terms, that knowledge archive supplies retention; the model of reality, the twins and the laboratories supply judgment; and the agents supply variation and critique.

Agent search can retain programs, revise its context or update its proposer. The label *swarm* does not specify which it does. The comparison in Table 2 isolates deployment with fixed proposer weights; EurekaZero aims to improve the proposer through training before a new problem arrives. Both need a reliable referee, and training costs belong in any account of deployment savings.

Table 2: **Where feedback can be retained.** The agent-search column describes fixed-weight deployment loops, not every agent system. The EurekaZero column states the proposed training and deployment protocol. In every column, people still set the aim.

Aspect	Human science	Agent search	EurekaZero
Aim	The researcher chooses	A person writes the goal	Problem encoding and metric, chosen by people for now
Variation	The scientist's ideas	Agents propose and critique	One candidate per test problem
Judgment	Experiments, peer review	Experiments, simulators, autonomous labs	Referee on hidden folds
Retention	The literature	Logs, archives, a map of knowledge	Incumbent pool; proposed writer updates
Where taste lives	Years of training, in one head	Weights fixed; context can adapt	Writer training (proposed)
Deployment budget	Varies by problem	Multiple adaptive attempts	One candidate; success measured

These examples separate retained solutions, task-specific selection and proposer updates. Section 3 traces how earlier AI methods connect those roles to executable tests.

3 The Road to Zero: Five Building Blocks

Five lines of AI research connect the four pillars of Section 2: deep learning, program search, reinforcement learning, games and the race for self-improvement. We trace what each proposes, how it is tested and what carries forward to the next attempt.[2]

3.1 Deep learning gave variation a voice

Deep learning built the variation pillar: machines that propose. Models learned representations from data [59]. AlexNet made the case on images [54], the Transformer modeled sequences with attention [112], and empirical scaling laws related language-model loss to model size, dataset size and training compute [42, 48]. Sutton [108] drew the lesson: general methods that use computation win.

Generative models turned representations into proposals in text, video [14] and code. Codex solved 28.8% of the HumanEval programming problems with one sample, and 70.2% with 100 samples, counting any that passed the unit tests [19]. That gap illustrates the value of a larger sampling budget. It is not a direct estimate of the geometric model in Equation (1): pass rates and the probability of beating an incumbent measure different outcomes.

Learned prediction and selection also support scientific discovery: protein structures that rival experiment in most CASP14 cases [47], 2.2 million crystal structures predicted

[2] Each history keeps only the milestones the argument needs; Appendix A collects the referees those milestones rely on.

stable relative to the Materials Project [68], and halicin, an antibiotic that then treated infections in mice [106].

CLIP put images and captions in one shared space [88], an analogy for our problem–program ranking map (Section 6). Reference training supervises the map with programs scored within each problem. Its ranks can influence search-parent selection (Section 6.4); they neither condition nor screen the writer. Map-ranking accuracy and first-proposal quality require separate tests.

Representations describe a problem, code generation supplies candidates and a separate map can rank drafts for search. Executable tests make their quality observable. EurekaZero proposes training on those outcomes, then measuring transfer beyond its training tasks.

3.2 Program search built the variation–judgment loop

Program evolution joined variation to judgment: propose a program, score it, keep the better ones. Friedberg [35] let a machine vary its own instructions, Holland [43] made variation and selection a general method of adaptation, and Lenat's AM and EURISKO searched concepts and heuristics by measures of interest [63, 64]. EURISKO takes its name from the Greek verb behind *eureka*, as ours does. Koza [53] closed the loop with genetic programming: a population of programs, a fitness function as referee and selection as coach. Populations retain programs; learned proposers can also retain patterns for later tasks.

Later work sharpened both halves: superoptimization [67], program synthesis from examples [38] guided by neural networks [5], neural architecture search [124], and AutoML-Zero, which evolved learning algorithms from basic mathematical operations [89].

Then language models became the variation operator. FunSearch found cap-set constructions beyond the best known [91]: verified new solutions within a fixed task, bought with search. Lion, an optimizer found by program search, went into production at Google [20]. AlphaEvolve multiplied two 4×4 complex matrices with 48 scalar multiplications, improving on Strassen's algorithm in that setting after 56 years [75]. MLEvolve carries the loop into machine-learning engineering [30].

EurekaZero adopts executable programs as moves inside this generate–judge loop, and asks how much of a verified search can transfer into the proposer itself. DreamCoder already learns reusable libraries and a search-guiding model across tasks [31]; a single-candidate evaluation tests how useful a learned proposal is before additional task-specific search.

3.3 Reinforcement learning turned retention into taste

Program search retains solutions; policy updates change the proposer. Reinforcement learning uses the referee's verdicts for those updates. Bellman [6] gave it an equation of value. Samuel's checkers program improved by playing a copy of itself [93], TD-Gammon reached near the best human backgammon players by self-play [109], and DQN learned Atari games from pixels [70]. AlphaGo learned from human games and searched at every move [100]; AlphaGo Zero learned by self-play alone [101]. AlphaZero mastered three games with one algorithm [102], MuZero learned a model for planning [95], and AlphaStar's league of mutually exploiting agents reached Grandmaster level in StarCraft II [113].

Box 2: Self-play in 90 seconds

In AlphaGo Zero, a player starts with the rules and plays itself. A short search before each move finds better moves than instinct would, and the referee scores each finished game. The network learns to propose the moves the search chose (its policy) and to predict the winner (its value), so the next search is stronger. Over millions of games, search is distilled into intuition [1, 101]. *What if the board were a problem, and the move an invention?*

In expert iteration [1], search is the teacher and the network is the student (Box 2). Language models also learn from feedback: human preferences [22, 78] and rule-based checks of answers, as in DeepSeek-R1 [39].

This line of work motivates training on referee-scored play. For EurekaZero, the next step is to define a game whose reward measures relative algorithm quality, then distinguish gains on training games from transfer to unseen families.

3.4 Gamification made judgment cheap

Games built the judgment pillar at scale. Shannon [96] chose chess because its moves and its goal are sharply defined. The common task framework made machine-learning research a game: shared data, enrolled competitors and a referee who scores every entry on a sealed test set [29], as in the Netflix Prize [7] and ImageNet [92]. People played science too. Foldit players beat automated methods on hard protein puzzles [23] and solved a protease structure that earlier attempts had not [50]; EteRNA players, whose designs were tested in a laboratory, found RNA design rules [60].

Then machines played problems. AlphaTensor's TensorGame beat the best-known matrix-multiplication algorithms for several sizes [33], and AlphaDev's AssemblyGame found sorting routines now in the LLVM C++ library [66]. AlphaGeometry learned olympiad geometry without human demonstrations [110]. AlphaProof, using the Lean proof checker, and AlphaGeometry 2 jointly reached silver-medal standard on the 2024 International Mathematical Olympiad problems, using computation over multiple days [45].

These benchmarks supply the recipe for a game (clear rules, a referee, a score and a sealed test) and raise the question of which parts of a task can vary under a shared representation and referee. EurekaZero makes the problem an input to the proposer (Section 4); whether that supports transfer across families remains a result to establish.

3.5 The race for self-improvement scaled search

The race for self-improvement spans program search, agent scaffolds and learned policies. Good [36] foresaw an

Table 3: **Related approaches retain experience in different forms.** This qualitative comparison separates output, deployment budget and retained state. Exact budgets depend on each study's protocol. The last row states EurekaZero's design; its proposal quality is still to be tested.

System	Output	Deployment budget	Retained experience	Feedback source
Genetic programming; AutoML-Zero	An algorithm	Study-specific search	Selected programs	Executable objective
AlphaTensor; AlphaDev	An algorithm for one family	Task-specific search	A policy for that game	Game reward
FunSearch; AlphaEvolve	An algorithm	Iterative search	Program archive	Automated evaluation
Research swarms (AI Scientist, autoresearch)	Experiments and code	Adaptive attempts	Context, logs and archives	Experimental outcomes
EvoTune; TTT-Discover; Evolution Fine-Tuning	An algorithm or construction	Protocol-specific search	Fine-tuned policy or search state	Search trajectories
TabPFN; MotherNet; KumoRFM	Predictions or network weights	Task-specific inference	Pretrained parameters	Training across tasks
Absolute Zero; STP	Answers and proofs	Decoding budget varies	Policy parameters	Correctness feedback
EurekaZero	An algorithm	One candidate	Proposed writer updates	Referee scores ranked within each problem

"intelligence explosion", and Schmidhuber [94] specified a Gödel machine that rewrites its own code once it proves the rewrite useful.

STOP improves the scaffold that calls a language model, including itself, and its authors note: "Since the language models themselves are not altered, this is not full recursive self-improvement" [120]. The Darwin Gödel Machine's self-editing agents raised its SWE-bench score from 20.0% to 50.0% [121], and AlphaEvolve cut Gemini's training time by 1% [75].

Language models also play themselves. SPIN plays its previous iteration [21], Absolute Zero proposes and solves coding tasks under a code executor [122], and STP pairs a conjecturer with a Lean prover [28]. Bailey et al. [3] attribute plateaus in LLM self-play to reward hacking by the conjecturer and evaluate self-guidance in formal theorem proving. Our closest neighbors learn to discover. EvoTune fine-tunes on what its own evolutionary search finds [107], and TTT-Discover trains on the one problem at hand, at a few hundred dollars per problem [119]. Evolution Fine-Tuning learns from 156,000 search trajectories across 371 tasks [61], and DiscoRL meta-learned a reinforcement-learning rule that beat human-designed ones [77]. These approaches retain improvements in different places: a search trajectory, a task-specific learner or a reusable learning rule. Our comparison asks about the quality and cost of the first complete candidate, rather than treating these forms of learning as absent.

Research itself is becoming a swarm. The AI Scientist automates ideas, experiments and papers [65], and the AI co-scientist ranks agents' hypotheses in an Elo tournament [37]. Karpathy's autoresearch, released in March 2026, runs about 100 experiments overnight and keeps what improves [49]. In May 2026, Recursive raised $650 million to build AI that improves itself [13]. In the innovation game, an encoded research task would be the board and its evaluator the referee (Section 4).

Self-improvement comes in three strengths: a system can find better external algorithms, build better components of itself, or improve its own ability to improve, raising verified discoveries per unit of compute across generations. Each requires separate evidence (Section 8); a recorded training trajectory alone establishes none of them on an unseen task.

These systems motivate two reference requirements: verify an improvement before retaining it, and sandbox candidate execution (Section 6.6). The proposed learning test compares a trained writer with its frozen reference under the same deployment budget; it is separate from the adapter record in Section 6.8.

3.6 Where the roads meet

Table 3 separates outputs, budgets and retained experience. The cross-task predictors and network generators [34, 44, 73] differ from readable-program output; one candidate does not equalize compute or inputs. Absolute Zero and STP use correctness feedback, whereas a declared incumbent sets our comparison standard; Section 4 defines how the referee judges a program against it.

Key idea

The test is whether experience on earlier discovery tasks improves the first complete proposal on a new one.

4 Innovation Is a Game

To make the comparison in Section 3 precise, we define one attempt at algorithm invention as a game: the encoded problem is the board, a program is the move, the referee checks and scores it, and a declared incumbent is the opponent. The

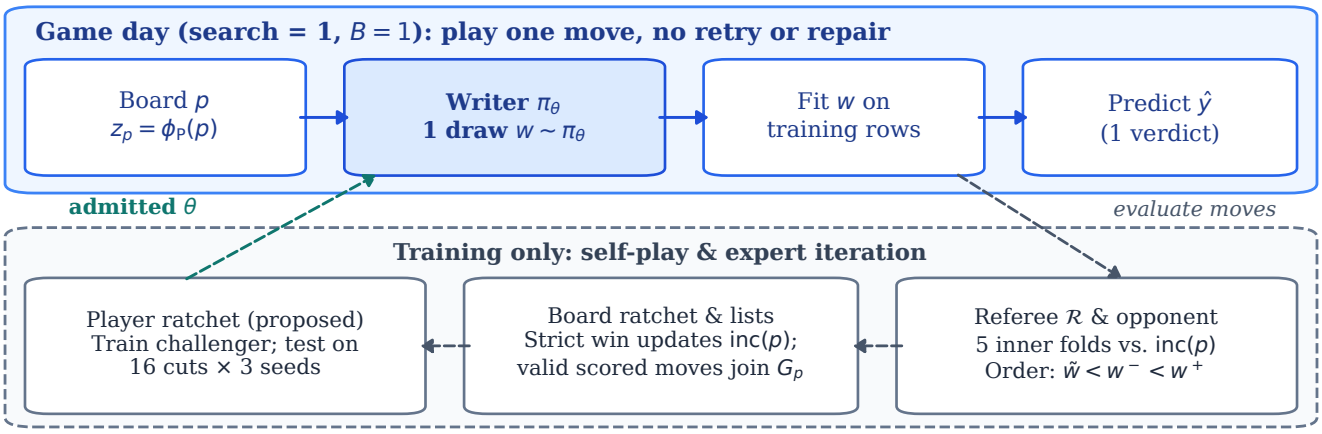


Figure 3: **Training plays the whole game; game day plays one move.** Reference loop (Section 6.5), separate from the adapter record (Section 6.8). Blue: game day; dashed: proposed training. Valid scored programs enter learning lists; failed attempts consume budget.

encoding supplies a shared interface; each task type still needs its own qualified encoding, language and evaluation protocol (Section 4.5).

4.1 Why games came first

Games became AI's laboratory because they supply four things at once: fixed rules, an impartial referee, a clear winner and cheap replay. Chess was chosen because it is "sharply defined both in allowed operations (the moves) and in the ultimate goal (checkmate)" [96], and checkers showed that a program could improve by playing itself [93].

Together these properties support repeated, reproducible evaluation. People scale judgment the same way, through prizes, leaderboards and the common task framework's "scoring referee" [29] (Section 3). Invention has the first three properties once the problem and the solution are written down, within the limits of Section 4.6; cheap replay is a question of cost (Section 5).

4.2 The rules

Table 4 sets the game beside Go. An illegal move is counted as a failed attempt. The reference writer π_θ reads the encoded problem. The referee keeps the incumbent as a separate comparison standard (Section 6.2); its code and score must stay outside the writer's input.

Formally, the game is a tuple of seven parts,

$$\mathcal{G} = (\mathcal{P}, \phi_P, \mathcal{S}, \phi_S, \mathcal{R}, r, \mathcal{O}).$$

In the reference learner, ϕ_P represents a problem $p \in \mathcal{P}$ as z_p, while ϕ_S embeds a program $w \in \mathcal{S}$ as z_w for the separate ranking map (Section 6.4). Game-day generation and the referee do not use ϕ_S. The referee $\mathcal{R}$ reports validity and score, the reward r orders moves, and the opponent $\mathcal{O}$ supplies the incumbent inc(p). The tuple extends Rice's model of algorithm selection [90]: the player writes the algorithm instead of picking it, and a declared incumbent sets the comparison standard.

Table 4: **Go and the innovation game, element by element.** Moves differ most: Go's are cheap, while each program is costly to judge (Section 4.6). Go's game length is from Silver et al. [100].

Element	Go	The innovation game
Board	Stones on a 19×19 grid	The encoded problem $z_p = \phi_P(p)$ for $p \in \mathcal{P}$
Move	Place one stone, or pass	Write one complete program $w \in \mathcal{S}$
Legal move	Empty point; no suicide or forbidden repetition	Parses, type-checks, runs within limits
Illegal move	Not allowed	Stillborn ($\tilde{w}$): an automatic loss
Referee	The rules of Go	The referee $\mathcal{R}$, on hidden folds
Win	Higher final score	Valid move w^+ beating inc(p) under r
Opponent	A past self	The incumbent inc(p) from $\mathcal{O}$, often a past self
Rating	Elo	Win rate and Elo across problems
Moves per game	About 150, each cheap	One (B), costly to judge

With one move, the game is a contextual bandit: the board supplies context, the program is the action and the verdict is the reward [57]. The incumbent supplies a changing comparison standard as verified programs displace it. The asymmetry is in the moves: Go offers about 250 legal moves per turn over about 150 turns, each free to check [100], while this game offers one move, costly to judge. Under a single-attempt budget, the chance q of a win matters directly (Equation (1)). Whether training raises that chance on new families is the test in Section 7.

Figure 3 separates training from game day. Its top row follows the single-candidate contract in Box 1.

4.3 The board and the move

The board is the aim pillar of Section 2, made machine-readable: a summary of the training data, the objective and metric, and the constraints. Landmark scores from cheap learners [82] and learned tabular summaries are possible encodings; Section 6.2 specifies the reference encoder and its cost limit. In the inspected implementation, that encoder consumes prepared numeric training rows and targets; the writer's engine and regression-task tokens are supplied separately. The objective, metric and budget are protocol settings, not additional fields in this encoder call.

A useful board should preserve distinctions relevant to algorithm choice, remain stable under irrelevant reorderings, exclude final evaluation information and have a measured cost. These are requirements to test, not properties guaranteed by naming an encoder. People still choose the problem and define what counts as a useful solution.

A move is one program in a typed language over learning primitives. A task-specific grammar restricts legal moves; Section 6.3 specifies the reference vocabulary and composition limits.

Types remove whole classes of stillborn programs before they run [71]. Canonicalization can merge syntactic variants; it need not identify every functionally equivalent implementation. AutoML-Zero reused scores when hashes of truncated predictions matched after a short training and validation probe [89]. Matching a probe does not establish agreement on other inputs (Section 6.3).

The language bounds what can be expressed; changing its primitives changes that space [9]. Structural constraints can prevent invalid programs; their effect on useful proposal quality must be measured (Section 7).

4.4 The referee and the opponent

The reward is an ordering first and a number second. Validity is a gate: any stillborn program scores below any valid one. Past the gate, programs rank by improvement over the incumbent on hidden folds. The reference promotion rule keeps the existing incumbent on a tie. Novelty never pays for losing: any novelty bonus (Section 5) must respect the order. For a stillborn program $\tilde{w}$, an also-ran w^- and a win w^+ on board p,

$$r(p,\tilde{w}) \;<\; r(p,w^-) \;<\; r(p,w^+).$$

The opponent is a ratcheting incumbent. A program must strictly improve under the same evaluation protocol to displace a board's incumbent. The reference design gives a retrained writer a separate proposed admission test (Section 6.5). The two rules define its self-play: a proposer faces incumbents retained from earlier play. AlphaGo Zero likewise separated network admission from move selection [101]. These are reference rules, not a record of writer improvement. The later loop in Section 6.8 updates only a ranking adapter; its default admission gate can retain equal utility. That record does not establish retraining or admission of the neural writer.

Subtracting the incumbent's score does not change which move is best. Promotion raises the bar, turns yesterday's winners into today's also-rans and shifts which problems are worth playing. The reference learning targets use each program's rank within its scored list (Sections 6.4 and 6.5).

4.5 One game, many boards

The accepted checkpoint route and both fixed comparators support only tabular regression. Binary and multiclass classification, ranking, forecasting and relational prediction remain prospective boards, each needing a qualified encoding, program language, metric and validation protocol.

Beyond machine learning, examples of referees include sorting tests [66], evaluators for mathematical constructions [91] and accelerator-kernel timers [75]. These motivate extensions of the recipe, rather than results of this paper (Section 8).

Casting algorithm discovery as a game is not new, but earlier games wrote one family of problems into their rules, as AlphaTensor did for matrix multiplication and AlphaDev for sorting [33, 66]. AlphaZero used one algorithm for three games and trained a separate network for each [102]. Here the problem is an input on the board, so one player can, in principle, learn many boards.

Key idea

A reliable referee makes a discipline a candidate for this game; transfer to that discipline still needs evidence.

4.6 The analogy's limits

A worked game, chosen by a rule fixed before inspection, is to be reported with the results (Section 7.6). The analogy breaks in four places.

The referee is noisy: picking a winner by a noisy validation score biases that score upward [17]. Strict numerical improvement alone does not quantify confidence; independent evaluation and an uncertainty protocol are needed to interpret it. **Moves are expensive**, each a fit and a cross-validation (Section 5). **Judges can be gamed**: the evaluation harness must resist exploits [114]. Section 6.6 specifies the isolation and parity checks required here. And **no algorithm wins everywhere**: averaged uniformly over all problems, no learner has lower error off the training set than another, ours included [118].

Real problems are structured, though, and a player that reads the board can choose its bias board by board. Section 7 reserves the tests of that hypothesis against fixed frontier-designed regressors, with paired outcomes beside Elo summaries [32]. Non-transitive comparisons are one reason not to reduce the evidence to a single rating [4].

The referee judges each attempt; cost accounting must also include invalid programs and valid attempts that lose. Section 5 prices the full sequence needed for a verified improvement.

5 The Price of Play

A verified improvement carries the cost of the attempts that precede it. The ledger includes encoding, input tokens, generation, repairs, referee execution, updates and operating costs. The planning ratios below organize that bill; a full-run price requires the measurements in Sections 5.6 and 7.

5.1 The bill in one line

For independent attempts with fixed win probability and cost distribution, expected cost per win is mean cost per attempt divided by that probability (Equation (1)):

$$C_{\text{win}} = \frac{C_{\text{attempt}}}{q}. \tag{2}$$

C_{attempt} is the mean incurred cost per candidate attempt, including failures. A repair that emits a revised candidate is another attempt; charge its work once. q is the probability that an attempt beats the incumbent.

Write $q = q_{\text{run}}\, q_{\text{beat}}$, where q_{run} is the share of protocol-valid attempts and q_{beat} the win rate among them. Report both with the overall win rate; q_{beat} is undefined if no attempt is valid. Valid ties and losses still cost money. Grammar constraints or selection using revealed outcomes can alter the validity rate without weight updates (Sections 6.3 and 6.8). These rates alone do not identify a learned improvement (Section 7.5).

Generating a token costs about 2 FLOPs per active parameter, and training on one about 6 [48]. Cost projections require reproducible inputs and calculations (Section 5.6).

For an observed batch, cost per verified win is the total ledger divided by verified wins, including update and operating costs where applicable. This accounting does not require independent attempts. Adaptive search still needs empirical cost and attempt curves; a batch with no verified win has no finite observed cost per win.

Figure 18 pairs program-event counts with full round durations in a later capture. That clock includes all work and waits; turning it into a proposal price still requires the attempt ledger.

Training and deployment need separate ledgers. Training pays for every attempt, including failures. The proposed deployment protocol pays for encoding, one proposal and its fit, with no repair. Fixed-weight agent loops pay for evaluation during task-specific search. Training a proposer may shift some of that cost to earlier tasks; the crossover still needs measurement (Section 7).

5.2 Wasted moves

The ledger distinguishes stillborn attempts from valid losses. A stillborn attempt fails to produce a usable program under the declared protocol: generation is incomplete, parsing or type-checking fails, execution exceeds its limits or crashes, or outputs fail predeclared validity checks. An also-ran is valid but does not beat the incumbent.

The pilot funnel planned for Section 7 (Figure 11, panel a) traces those stages and the compute spent before each failed attempt stops.

If 90% of attempts fail to improve the incumbent, the independent-try model requires ten attempts per improvement in expectation. A typed language can exclude invalid structures before fitting (Section 4). Runtime, resource and output failures may consume part or all of the fit, and their incurred cost must be counted. The pilot must measure whether also-rans dominate. The matched writer comparison in Section 7.2 holds the budget, incumbent and referee fixed.

5.3 The correction tax and the reasoning dial

Here and in Section 5.6, we consider language-model writers for regression and the agent comparators of Sections 7.3 and 7.4. The compact reference network and its generation limit are specified in Section 6; the scaling examples here concern language-model writers.

Small models are cheap per token and can be dear per win. For generation alone, compute per token falls about 190-fold from a hypothetical dense 1.5T writer to an 8B one. With equal token counts and compute unit costs, the smaller writer can tolerate a win rate about 190 times lower before that component's cost per win matches.

If it needs three times the tokens per attempt, the generation-only ratio falls to about 60. Repair turns also re-read a growing context. Encoding and referee costs change the full-cost ratio; a threshold for the complete bill in Equation (2) requires counting every component.

Reasoning is the second dial. Reasoning tokens may cut stillborn moves, but each is billed as output. The planned pilot (Figure 11, panel b) compares cost per win over writer size and reasoning effort. A U-shaped curve is the hypothesis; the measurement decides where, and whether, the minimum sits.

5.4 Rewarding useful hallucination

A familiar solution can be a useful prior. Matching the incumbent's score earns a tie; promotion requires improvement under the same protocol. A *useful hallucination* here is an unfamiliar proposal whose value the referee verifies (Box 3). Possible exploration controls include temperature and entropy schedules, a curriculum past known solutions, and quality-diversity archives [62, 72]. Any novelty bonus must preserve the reward ordering in Section 4. These are design options.

The reference training routes combine policy draws with grammar-based fallback and random exploration to seed verified search (Section 6.5). An exploration study must declare those controls and their budget shares, including unsuccessful attempts; Section 7 reserves the quality–cost comparison.

Box 3: What a referee verifies

An unfamiliar program is a conjecture; the referee checks whether it runs and improves the declared score. Verification establishes that bounded result. A claim of discovery also needs comparison with known methods (Section 8). The Go example in Section 2.4 illustrates this distinction. The referee's reliability and execution cost must be established for the task (Section 6.6).

5.5 How many games?

Keep three quantities separate. **Tries to find one improvement:** $1/q$ on average (Equation (1)). For independent tries with fixed $q = 1\%$, 69 attempts reach even odds and 299 reach a 95% probability of at least one win in that model. Report empirical attempt curves for adaptive or correlated proposals.

Experience to train a transferable player: AlphaGo Zero generated 4.9 million self-play games in about three days [101], and AlphaZero played 44 million games of chess in about nine hours [102]. An illustrative grid of 10,000 problems, 100 rounds and 8 attempts per problem per round gives 8 million attempts, the same order. The grid counts work, not learning: the number of problem families matters more, and pilot learning curves, read only within their measured range, must decide.

Precision to plan the test: illustratively, 150 independent units at a win rate near 60% give an approximate 95% interval of ± 8 points. The actual study must compute uncertainty over independent families, rather than treating every dataset as an independent observation.

5.6 Pricing evidence still required

Turning the per-token scaling comparisons of Section 5.3 into a hardware bill requires accounting for resident model memory alongside active compute. Mixture-of-experts writers use compute on active parameters but need memory for all parameters: Kimi K2 has 32B active of 1T, while DeepSeek-V3 has 37B active of 671B [27, 51]. Resident one-byte weights for 1T parameters alone require at least 13 H100 SXM GPUs with 80 GB each [76]; runtime state needs additional memory. This memory floor is separate from the full cost projection discussed next.

A reproducible projection therefore needs a pinned model and hardware configuration, dated prices, measured or justified utilization and an explicit accounting boundary. Those inputs remain unrecorded; a full cost projection remains a measurement task.

Key idea

Cost per verified improvement depends on both the cost of an attempt and the probability that it wins.

Under Equation (2)'s fixed-probability model, doubling q halves expected cost per win only if mean attempt cost stays fixed. Measuring that tradeoff requires a common task mix, incumbent and verification rule.

6 EurekaZero: Learning Taste

The reference design specifies training a writer and separate ranking map on search-scored lists, then deploying the frozen writer for one proposal. The record in Section 6.8 updates only a ranking adapter over heuristic, retrieval and grammar proposals. Their cost ledgers (Section 5) and evaluations are separate.

6.1 The player in one picture

Figure 4 connects the architecture to the contract in Box 1: a frozen encoder reads the board and π_θ writes a complete program, as AlphaGo's policy proposed a move [100]. In one reference search route, the separate map v_ψ supplies weights for parent selection among ranked survivors (Section 6.4); it neither conditions nor screens the writer.

6.2 Reading the board

The reference regression encoder turns the board of Section 4 into 2,048 numbers, $z_p = \phi_{\mathrm{P}}(p)$, in three blocks. The first summarizes a pretrained row encoder's embeddings across rows, independent of row order (512 values). The second projects the hidden state of a pretrained third-party tabular regressor that reads the training rows and labels (1,024; third-party provenance pending; restricted; redistribution prohibited). The third summarizes two 32-element windows of training targets scaled to $[0, 1]$ (512 values; constant targets map to zero).

Preparation caps the combined tabular splits at 50,000 rows. Encoder inputs remain training-side only: the regressor block samples at most 768 context rows and 256 query rows from those training rows. Frozen meta-training cohort moments standardize each component block before concatenation and elementwise clipping to $[-32, 32]$.

The reference encoder is pinned by hash and stays fixed during writer training. Encoding is bounded at 30 seconds per problem. A context with any non-finite value is refused rather than imputed (Appendix C). The input contract permits training-side task evidence only; final held-out labels and their derived statistics must stay outside the input closure. A context's recorded declaration that no final target was used establishes neither input closure nor numerical qualification. CPU contexts require native replay and bounded-execution receipts before accepted-policy use.

Whether problems with similar best programs sit together requires a separate representation study; Section 7.5 reserves that test.

6.3 Describing the move

A move is one program in the typed language of Section 4, with a 401-token training inventory listing 31 transforms and 22 prediction heads; the product runtime restricts executable heads. The writer emits a flat operator-and-slot token stream with a fixed beginning, engine and regression-task header, rather than parenthesis tokens or the `pipe` wrapper. Its nine-phase grammar mask permits up to six transform stages and three expression stages; each expression tree is bounded at depth four and 15 nodes [24, 56]. After the end token, detokenization renders a parenthesized program that must parse before entering the pool. The mask has no remaining-token-budget check: a draw without an end token at the 128-token horizon, including the fixed header, is incomplete and discarded. Grammar legality does not establish successful execution or useful quality. The proposal pool removes exact decoded-string duplicates and records their SHA-256. The

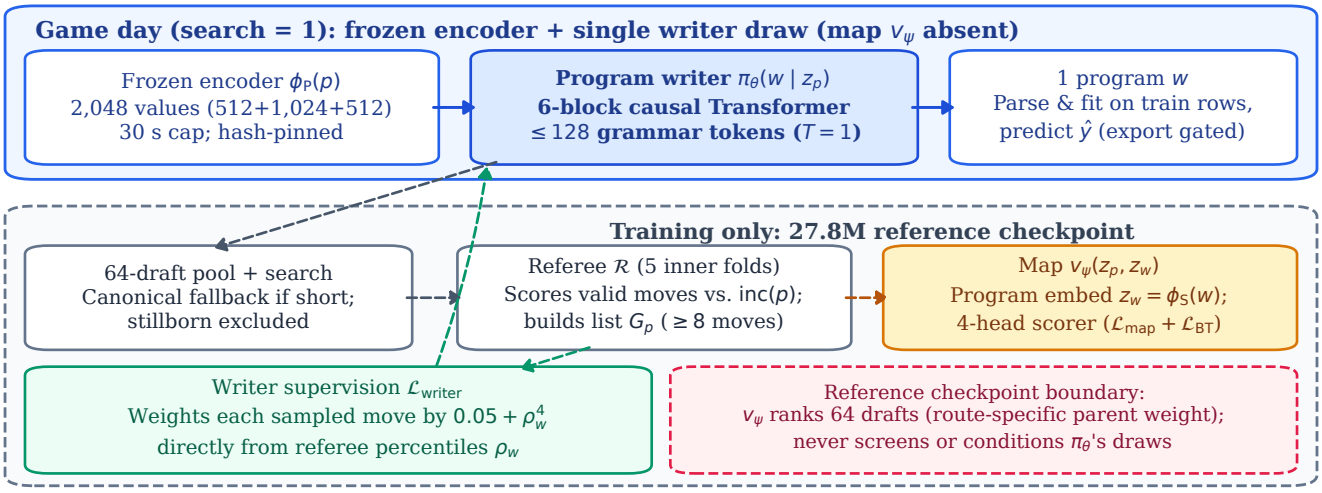


Figure 4: **Reference training design for the writer and separate ranking map.** Top: game day, the whole of search = 1. Dashed box: proposed training. Search extends the writer's drafts; referee-scored lists supply objectives for the writer (θ) and map (ψ). Stillborn programs enter no list.

executor also records canonical recipe hashes. Probe-based behavior matching is a separate audit, not part of pool deduplication. Without probe evidence, this behavior check reports novelty as unknown. Readable Python and Rust exports are a reference requirement. Legacy recipe emitters reject the synthesized input stage; the separate fitted-state path is allowlisted. Reported winners still require the parity check in Section 6.6.

6.4 One map for problems and programs

The map reads a program as $z_w = \phi_S(w)$, the layer-normalized mean of non-padding token embeddings, including `BOS`, the fixed engine and task header, and `EOS`. Pooling discards token order, limiting its ability to distinguish compositions built from the same tokens.

The map's own two-layer context network projects the 2,048-value z_p to $r_z \in \mathbb{R}^{512}$. It joins three 512-value blocks: z_w as is, $z_w \odot \tanh(W_\gamma r_z + b_\gamma)$, and $|z_w - (W_{proj} r_z + b_{proj})|$. Four two-layer heads receive this same 1,536-value interaction vector; their mean gives the ranking logit $v_\psi(z_p, z_w)$. Native ranking losses supervise that mean jointly, without separate targets or bootstrap samples. Training applies dropout of 0.05 in each ranking head. The runtime head instead reads $[r_z, z_w]$, while the program-only control reads z_w.

Higher logits rank higher within a problem. The score is not a calibrated error estimate. Comparing its magnitude across problems or reading it as a probability of a verified win requires separate evidence. The referee still establishes validity and measures performance.

The map's training objective uses scored lists G_p for problem p; $\rho_w \in [0,1]$ is a program's percentile, 1 for the best. Native training requires at least eight scored programs in a source list and samples eight per comparison: its champion and seven quantile-stratified comparators. The lower-level listwise loss accepts groups of at least three with finite percentile targets (Appendix C). The list-level form trains the map's softmax toward a target concentrated on the best [16]:

$$\mathcal{L}_{map}(p) = -\sum_{w \in G_p} \frac{e^{\rho_w/\tau}}{\sum_{u \in G_p} e^{\rho_u/\tau}} \log \frac{e^{v_\psi(z_p, z_w)}}{\sum_{u \in G_p} e^{v_\psi(z_p, z_u)}}, \quad (3)$$

with native target temperature $\tau = 0.25$. Temperature scales the target percentiles; the mean ranking logits enter the prediction softmax unscaled. With a unique best program, the low-temperature limit concentrates the target on that program. This is a listwise ranking objective over programs scored on the same problem; the analogy to contrastive learning [88, 111] does not establish CLIP-style alignment across problems and programs. A Bradley–Terry term adds pairwise comparisons on the same lists [12]. Stillborn moves have no score, so this ranking objective covers only programs that run.

The program-only head controls for ranking without problem context (Section 7.5). Ranking valid scored programs alone does not test the validity rate q_{run} in Section 5.

Key idea

The reference game-day protocol omits the ranking map. Matched first proposals test whether training improved the writer.

In the native implementation, the writer and map use separate context networks and token embeddings. The map ranks drafts without dropping them or screening the writer. One native GP route matches map-derived weights by program identity to ranked survivors for parent selection. The bounded search driver instead cycles among up to four parents chosen by measured loss. These are different influence paths, not evidence of trained-map lift; an evaluation must bind the route it used.

Only the 64 neural or fallback seeds receive normalized rank weights; the 16 tail seeds enter unscored. Unscored survivors share the lowest scored survivor weight while scored survivors remain. Native assembly takes up to ten same-cut teachers, then cross-cut teachers up to a 16-seed tail. Random fallbacks fill to ten slots if needed; random exploration fills the remaining slots. The standalone seed writer instead uses 16 random programs. Teacher-first execution prioritizes retained cut teachers; Algorithm 1 leaves the tail route-dependent. Search-parent weights and writer-loss weights act at different stages; the latter use referee rank. Whether map-guided targets improve later proposals requires the matched ablation in Section 7.5. The historical diagnostics in Figure 17 do not supply that comparison.

6.5 The self-play loop

The native proposal sampler seeks 64 distinct parseable programs at $T = 1$, scheduling at most eight batches of 128 neural draws (1,024 total). To fill a shortfall, it tries up to 10,000 seeded random canonical pipelines and projects them through the grammar. It refuses an incomplete pool. These are training-search budgets, separate from the one-draw protocol in Section 6.7.

The reference training rule is expert iteration [1]: in the proposed offline loop, search-scored programs train the writer. The writer term weights sampled programs from G_p by referee rank; it joins the map and auxiliary terms in the combined objective (Appendix C):

$$\mathscr{L}_{\text{writer}}(p) = -\sum_{w \in G_p} \left(0.05 + \rho_w^4\right) \log \pi_\theta(w \mid z_p), \qquad (4)$$

$$\mathscr{L} = \mathscr{L}_{\text{writer}} + 0.1\,\mathscr{L}_{\text{map}} + 0.25\,\mathscr{L}_{\text{BT}} + 0.1\,\mathscr{L}_{\text{runtime}} + 0.05\,\mathscr{L}_{\text{prog}}. \qquad (5)$$

The writer loss in Equation (4) is schematic: it displays the rank weights. The native implementation also smooths grammar-legal targets, averages token losses within each program, and normalizes by the sum of program weights (Appendix C).

Here $\mathscr{L}_{\text{BT}}$ is a Bradley–Terry loss on the same lists, $\mathscr{L}_{\text{runtime}}$ a Huber loss between $\ln(1+s)$ and $\ln(1+\text{softplus}(h))$, where s is nonnegative measured seconds and h is the raw runtime-head output, and $\mathscr{L}_{\text{prog}}$ the Bradley–Terry loss of the program-only control head. A median-ranked program carries about a tenth of the top-ranked program's weight in the writer loss. Its referee score also contributes to the map's ranking targets.

The reference procedure uses two ratchets (Section 4). A program displaces a board's incumbent only by strictly beating it.

The proposed reference admission test calls for 16 cuts with three seeds each. A challenger may replace the serving writer only if its mean log error ratio favors it. It must also win at least as many cuts as it loses and fail no more often. Here a cut is a held-out evaluation case fixed before the admission test. For each cut, the rule takes the natural log of paired best-error ratios (challenger divided by serving) and averages over matched seeds; it then averages across cuts. Negative values favor the challenger. This admission comparison is separate from the single-program comparison in Section 7.2.

The historical reference. The decoder-only writer has separate ranking and auxiliary heads. Appendix C records the architecture, registered capacity, token allocation and training corpus in one inventory.

These specifications do not supply the checkpoint's acceptance evidence or qualify its runtime. Both records remain inputs to the evaluation contract (Section 7.1). The accepted native checkpoint route is limited to tabular regression, and the native training entry point remains held at its grant gate. Numerical parity of the CPU float32 Candle port with the native CUDA BF16 tch runtime remains unqualified; random streams and uniform-token fallback differ.

6.6 Keeping the referee honest

Search supplies training targets only as reliably as its referee. A learner rewarded by a referee can exploit its flaws. In February 2025 Sakana AI walked back claims after its AI CUDA Engineer exploited the evaluation code [116]. In June 2025 METR reported frontier models patching their scorers [114], and 16% of 1,968 agent-benchmark tasks can be passed without being solved [123].

The evaluation protocol must withstand such attacks. Selection, promotion and final scoring must use separate data (inner folds, held-out cuts, a final fold scored once), with sealed meta-test families outside the learner's inputs.

The proposed comparisons require sandboxed execution using seccomp and Landlock, with memory, swap and time limits. A planted label leak and a sealed canary must both be caught. Every reported winner must be re-run from its emitted code, with native and emitted predictions agreeing within $10^{-9}\max(1, |\hat{y}|)$.

One earlier leaderboard entry for our search selected each dataset's best test-scored outcome from a ledger of 63,108 evaluations; we exclude it as test selection (Appendix B).

6.7 Game day: search = 1

Algorithm 2 applies the contract in Box 1 to regression: frozen components, a recorded seed and one draw, with failures and all incurred costs retained. Section 7.2 compares that draw with a frozen reference under matched conditions.

Report later search separately, as Silver et al. [101] reported AlphaGo Zero's raw network apart from its search. The adapter record below concerns candidate selection and admission; the matched first-proposal comparison remains the test in Section 7.2.

6.8 A separate adapter admission record

This is the adapter record (Figure 5) for the second learning task in Section 6. Utility is the logged scalar score, with larger values treated as better.

Algorithm 1 The reference self-play loop (training).

Require: training problems; serving writer π_θ and map v_ψ; incumbents $\mathrm{inc}(p)$

for each round **do**
  for each sampled training problem p **do**
    $z_p \leftarrow \phi_{\mathrm{P}}(p)$; keep the first 64 distinct valid draws from $\pi_\theta(\cdot \mid z_p)$ ▷ $T = 1$; bounded fallback if short
    verified search, seeded with the 64 drafts and 16 tail seeds (teachers/fallbacks and random exploration), proposes more programs ▷ the teacher
    $\mathscr{R}$ fits and scores each distinct program on 5 inner folds ▷ stillborn: no score
    if a program strictly beats $\mathrm{inc}(p)$ **then** it becomes $\mathrm{inc}(p)$ ▷ board ratchet
    add every scored program to the list G_p ▷ winners and also-rans
  train a challenger (θ', ψ') on the lists
  if it beats π_θ on held-out cuts **then** $(\theta, \psi) \leftarrow (\theta', \psi')$ ▷ proposed player ratchet

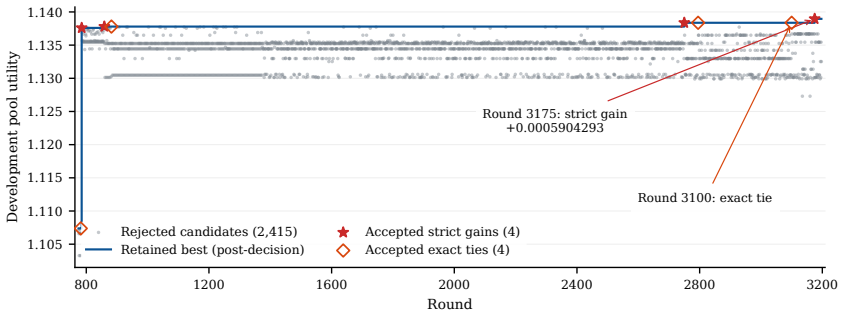


Figure 5: **Separate adapter admission trajectory.** The captured prefix in Table 10: blue is post-decision retained best; grey marks rejected candidates, red stars strict gains and hollow diamonds exact ties. Decisions use unrounded logged values. Scope: Section 6.8.

Algorithm 2 Proposed game-day protocol (search = 1).

Require: new regression problem p with training rows and metric; frozen ϕ_{P} and π_θ

$z_p \leftarrow \phi_{\mathrm{P}}(p)$ ▷ read the board
$w \sim \pi_\theta(\cdot \mid z_p)$, one draw at $T = 1$ ▷ one move; no retry
$f \leftarrow \mathrm{FIT}(w,$ training rows$)$ ▷ as any algorithm is fit
$\hat{y} \leftarrow f($new rows$)$ ▷ a stillborn w scores as a loss
return w, $\hat{y}$

Scope of the telemetry. All records here and in Appendix D are adaptive training-side observations. Program-incumbent events and adapter admissions have different denominators; neither establishes independent first-proposal improvement or deployment throughput. The aggregate records do not yet bind an exact loop source, adapter, evaluation population or normalization, so cross-model comparisons and transfer remain unqualified. Retained windows are separate; unobserved gaps are not filled or joined into a learning curve. Independent comparisons belong to Section 7.

Typed program edits, cross-family retrieval and grammar synthesis supply candidates in the inspected implementation.

The inspected loop freezes every network parameter, then enables updates only for non-anchor parameters of the cost-ranking adapter. Its anchor, encoder and other modules stay frozen. This loop has no neural program writer. Its pool-utility calculation uses the recorded scores and gate outcomes of the first four candidates as revealed history, then selects the lowest predicted cost from the pool, including those entries. Training puts the lowest recorded scores first in its slates; utility evaluation uses stored pool order. The calls can therefore reveal different examples.

The caller supplies modality tokens and a mask. It separately zeros the 584-entry context feature vector and both prior inputs, and supplies a fixed scale of 0.4208; zero context features do not mean absent modality tokens. Its objective combines regression to recorded costs and listwise matching; failed gates receive target cost $+2.0$. Reporting averages negative selected scores, with utility -2.0 for failed gates. The full-cohort route disables development-anchor margin projection; smaller-cohort runs can enable it. These routes require separate receipts.

Separate adapter ablations on historically seen training families found a benefit from the revealed-outcome kernel even at random initialization, with fewer invalid selections. This structural effect does not establish a gain from optimizer updates or from the separate reference map. A claim about learning therefore needs matched trained and untrained controls with the same candidate pool and revealed history, reporting invalid selections separately from quality among valid programs.

Candidate programs are evaluated on two training-side folds; the historical reference procedure specifies five. Evaluations are logged separately from training-pool admission: ordinary admission requires passing gates and a recorded score below the training context's incumbent by more than 10^{-6}. Historical programs have a separate admission path. The adapter accepts ties by default. Optional strict admission

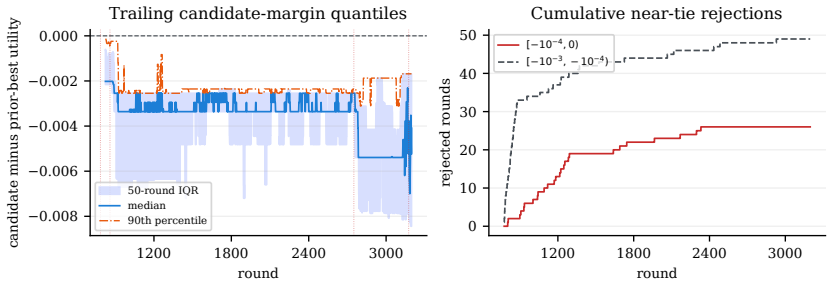


Figure 6: **Rolling candidate margins and near-tie rejections.** Left: trailing 50-round median, interquartile range and 90th percentile of candidate minus pre-decision retained-best utility, including admitted candidates. Quantiles begin at the first complete 50-round window; dotted vertical lines mark strict-gain rounds. Right: cumulative rejected rows in two exclusive margin bands. Both panels use the full prefix in Table 10; no unobserved rounds are inferred. Scope: Section 6.8.

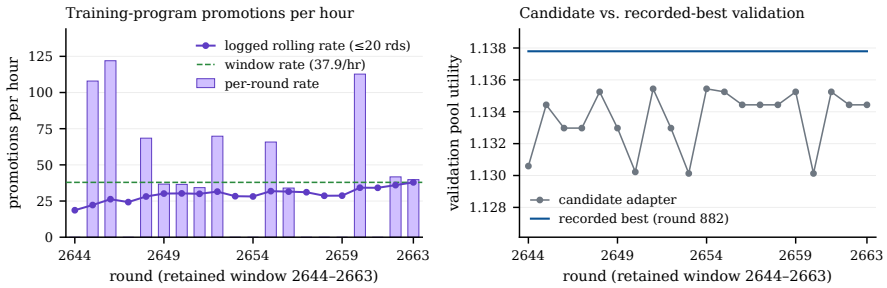


Figure 7: **Training-program promotions per hour and adapter validation trajectory.** Left: logged rolling and per-round strict training-program promotions per hour. Right: candidate and recorded-best adapter validation pool utility. Both panels cover the 20 consecutive completed rounds 2644–2663 (21 strict promotions over 1,994.54 s of logged wall time, or 37.90 promotions per hour, with zero adapter admissions). Scope: Section 6.8.

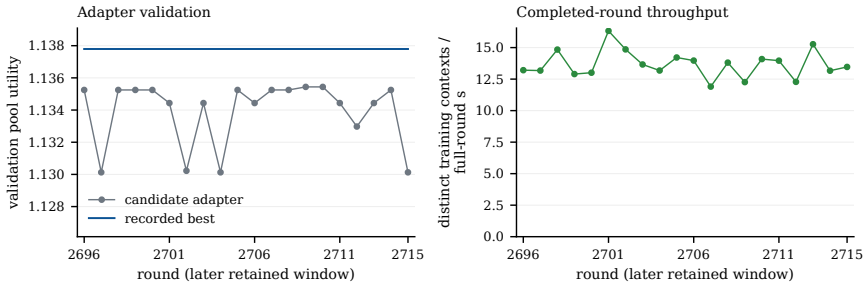


Figure 8: **Adapter utility and whole-round throughput in a later retained window.** Both panels cover the same 20 completed summaries, rounds 2696–2715. Left: candidate adapter validation utility and the recorded best. Right: distinct training contexts divided by the full wall time of each completed round (contexts per second). The rates include all work and waits inside that interval. Scope: Section 6.8.

requires utility above the retained best plus an explicit non-negative margin. Rejection restores the retained model and optimizer states. GP search continues, and the completed-round clock includes rejected adapter work.

[TBD: Bind the retained snapshot to the exact loop source, adapter, candidate pools, revealed-history inputs and utility normalization.]

The retained admission ledger. Table 10 is the canonical chronology and count table. Figures 5 and 6 show its trajectory and pre-decision margins. A recorded best-round identity can change on a tie without raising utility.

Figure 7 puts training-program incumbent events on a completed-round clock; Figure 8 divides distinct within-round training contexts by full round time. Their retained windows and denominators differ from the admission prefix.

The remaining windows and historical comparisons are in Appendices D, D.10, D.12, D.13, E and F. Their scope is the statement above; the matched first-proposal test is in Section 7.2.

7 Results

With the single-candidate budget fixed, the tests below ask whether training improves a first proposal on an unseen family. The development records in Section 6.8 concern program search and adapter admission; their clock and decisions do not establish deployment performance. Table 5 separates the

Table 5: **Evaluation plan at a glance.** Each test is specified in the subsection named in the first column. No outcome is reported yet; the contract in Section 7.1 fixes inputs, budgets and splits before outcomes are opened.

Test	Question	Comparison or control	Reported in	Status
The effect of training (Section 7.2)	Does training improve the first proposal on unseen families?	A frozen reference writer under the same budget, paired first draws	Text of Section 7.2	Pending
One proposal against fixed methods (Section 7.3)	Does one task-conditioned program compete with methods that run unchanged across tasks?	Frozen algorithms whose receipts record the designing model	Table 6 and Figure 9	Pending
The cost of more attempts (Section 7.4)	What do successive attempts buy, per attempt and per unit of cost?	Agents allowed successive attempts; the pilot cost funnel	Figures 10 and 11	Pending
What the map contributes (Section 7.5)	Does the map read the problem, and does it change the writer?	The program-only head; a matched causal ablation	Table 7	Pending
Failure cases and a worked example (Section 7.6)	How do attempts fail, and what does one game look like?	Terminal outcome classes; a rule-selected example	Table 8	Pending

training, fixed-method, search and map comparisons from the failure analyses. All first-proposal evaluation outcomes remain pending. **[TBD: Add independent evaluation evidence before making a performance claim.] [TBD: Supply a source-bound benchmark before claiming deployment throughput.]**

7.1 Evaluation contract

Freeze the writer, permitted task inputs, output support, decoding rule and budgets before opening final outcomes. Pin dataset versions and content hashes, row splits and family assignments. Separate unseen problem families from training and development, and report which task types were actually tested.

Report known training and pretraining exposure for the writer, encoders and learned comparators, with unresolved provenance marked explicitly. Bound unseen-family claims to the exclusions actually established.

Each EurekaZero submission follows Algorithm 2, counting failed generations and executions in the overall win rate. Also report validity and wins among valid attempts; neither rate alone establishes learning. Pair outcomes and uncertainty by family, with encoding, generation, fitting and prediction costs. Disclose internal model or hyperparameter selection within the emitted procedure. **[TBD: Document the final problem inventory, dataset identities, row splits, family assignments and exclusions, component provenance, evaluator, checkpoint, seeds, task metrics and failure accounting.]**

7.2 The effect of training

Use a frozen reference writer as the control. Predeclare its checkpoint and the evaluated training intervention before opening final outcomes. Hold the encoder, program grammar and output support, permitted task inputs, decoding rules and fitting protocol, referee, incumbent and resource budgets constant; vary the writer checkpoint. Changes to the support, encoder or evaluator belong in separate conditions.

Table 6: **Planned paired comparisons.** No outcomes are yet supplied. Report ties, failures and excluded cases alongside wins.

Comparator	Outcome and uncertainty
[TBD: first frozen comparator identity]	**[TBD: paired evaluation]**
[TBD: second frozen comparator identity]	**[TBD: paired evaluation]**

[TBD: Leaderboard: supply the evaluated methods, paired outcomes, rating protocol and uncertainty.]

Figure 9: **Leaderboard pending.** Elo is a summary of a specified comparison population, not a substitute for paired task results.

Pair the first draws under predeclared seeds and report all of them, with no selection of the best seed or candidate. Report paired family-level outcomes, failures and costs alongside source-bound evidence of optimizer updates and parameter changes. **[TBD: Supply the matched first-proposal comparison and training-update evidence before claiming transfer.]**

7.3 One proposal against fixed methods

[TBD: Identify Jepson's supplied frozen comparators, their provenance and eligibility, and the matched outcomes and uncertainty for Table 6 and Figure 9.]

7.4 The cost of more attempts

Compare the single proposal with agents allowed successive attempts and development feedback. Keep final outcomes unavailable to those agents. Count every candidate attempt, including incomplete or invalid outputs and failed executions; a repair that emits a revised candidate is another attempt. Charge all task-time computation.

[TBD: Search curve: performance versus attempts and compute, including failed attempts and unreached targets.]

Figure 10: **Deployment search comparison pending.** The single-candidate condition fixes a budget; its measured quality remains to be established.

[TBD: Pilot cost measurements: supply per-stage failure shares and cost per verified win.]

Figure 11: **Pilot cost measurements pending.** Planned panels: (a) the share of development proposals that run, score, beat the incumbent and survive re-verification, with the compute spent at each stage; (b) cost per verified win by writer size and reasoning effort. **[TBD: Supply pilot data, seeds and family-level uncertainty; identify settings with no verified win without assigning a finite cost per win.]**

Table 7: **Map evidence still required.** Ranking quality and a change in first-proposal quality are separate outcomes.

Question	Required evidence
Does the map rank candidates?	**[TBD: held-out ranking study]**
Does it improve the writer?	**[TBD: matched causal ablation]**
Does it reduce execution failures?	**[TBD: failure-class comparison]**

Plot performance against both attempts and cost (Figure 10); if an agent does not reach a reference performance within its budget, report that limit. **[TBD: Supply attempt curves, matched budgets, repeated seeds and uncertainty.]** Pilot cost measurements from Sections 5.2 and 5.3 are planned for Figure 11.

7.5 What the map contributes

Test whether nearby problem encodings share useful programs, using data disjoint from the data that fit the map. Compare ranking agreement with the program-only head, so that a global preference for common programs can be distinguished from reading the problem. **[TBD: Supply problem-similarity and program-only-control evaluations.]**

Agreement with the referee tests ranking, not improvement of the writer (Table 7). The reference map neither conditions nor screens a writer draw (Section 6.4); its accuracy on valid scored lists cannot establish a reduction in invalid proposals or an overall win-rate gain.

For a causal test, bind the parent-selection route and evaluated checkpoint. Vary the map's influence while holding the initial writer, candidate-generation rules, training budget and deployment protocol fixed. Measure the resulting writers' paired first proposals, including validity, wins and incurred costs. **[TBD: Identify an implemented influence path or limit the study to ranking.]**

Table 8: **Worked-game record.** Fields reported for the game selected by the rule above. An empty selected outcome category is reported rather than replaced.

Field	Record
Problem, task and metric	**[TBD: Identify the problem, task and metric.]**
Encoding time	**[TBD: time]**
Emitted output	**[TBD: program or unfinished draft and size]**
Check and fitting outcomes	**[TBD: outcomes and time]**
Hidden-fold score of a valid fit	**[TBD: score]**
Incumbent's score	**[TBD: score]**
If no valid fit	The failure and its incurred cost, in place of the scores

7.6 Failure cases and a worked example

Describe unsuccessful proposals and coverage limits as well as wins. Assign each attempt one terminal status: incomplete generation, parse or type-checking refusal, execution or resource failure, invalid output under declared checks, and valid ties, losses and wins. Charge every attempted candidate its incurred cost. Unresolved outcomes retain an explicit unknown status; fix their treatment and the validity rules before opening final outcomes. **[TBD: Supply attempt inventories and costs by outcome class, identifying unresolved cases.]**

Select the worked example by a rule fixed before inspecting outcomes. Report its emitted program, fitting procedure and comparison outcome. If the selected category is empty, report that fact. **[TBD: Supply the example selection rule, actual program and verified outcomes, with no replacement of failed attempts.]** Table 8 lists the record for the selected game. The attempt permits no repair or second try, and its sealed verdict is reported without updating the writer.

7.7 Earlier development diagnostics

Closed-out development studies evaluate a frozen reference adapter and its modifications, rather than the writer's independent first proposals. The first three studies use 1,219 development-training contexts from 205 families and three seeds with different five-way family partitions. Prior fitting and gradient-clipping continuations exclude evaluation and selection families. Test-time adaptation uses only the four revealed candidates in each context and resets its weights after scoring. Fits or settings were locked before outer-fold evaluation. Contexts are averaged within families; families receive equal weight. Utility-gain intervals use 10,000 paired-family bootstrap draws shared across seeds and summarize the median seed gain. Reported intervals are 95%; p values are one-sided, with one primary contrast per study. These studies reuse a development population; they are not independent replications.

Calibrated blend. A calibrated tree prior receives weight 0.25 in a cost-space blend with the reference adapter. Its primary endpoint is the maximum utility among four revealed

and four lowest-cost unrevealed candidates, rather than a single native choice. The gain is $+0.0137$ (95% interval $[+0.0056, +0.0259]$; $p = 0.0005$ against a $+0.001$ margin). Elo is 1523.1 $[1513.5, 1532.4]$, relative to the reference at 1500. Family harm below -0.01 is 7.3% (15/205). The result is a **candidate**, with no adoption and zero adapter optimizer updates; the tree prior was fitted.

Per-context clipping. Clipping each context's gradient before averaging, with 40 updates per trajectory, gives utility gain -0.02510 ($[-0.05128, +0.00009]$; $p = 0.9743$; 0/3 positive seeds). Elo is 1485.63 (lift -14.37, 95% interval $[-24.34, -4.38]$). Family harm below zero is 46.83%, and the treatment does not strictly beat its shuffled-factor placebo: **NO_ADVANCE**.

Revealed-only adaptation. Selection on a separate panel chose one SGD step at learning rate 10^{-5} in all 15/15 cells. At that locked setting, adaptation changes scores in 1,219/1,219 contexts but changes no selected program. The largest grid budgets flip only 1–4 of 1,219 choices. Utility gain is 0.000000, with interval $[0.000000, 0.000000]$. Elo is 1500.00, with interval $[1500.00, 1500.00]$. The verdict is **NO_ADVANCE (inert treatment)** within the tested one-to-five-step, 10^{-5}–10^{-4} learning-rate grid; it does not exclude effective adaptation at larger budgets.

Admission audit. The sole primary attempt is **INCOMPLETE**, with formal verdict **INCONCLUSIVE**: one of 91 choices in 19 development-validation families differs from its archived choice, failing the native parity gate. Post-failure descriptive replay of a stricter admission rule accepts 0/2,418 historical proposals. This replay neither qualifies the checkpoint nor establishes future power or sequential error control.

The single-choice studies use the selected candidate's utility, with invalid choices assigned -2. Their endpoint differs from the blend's best-of-eight endpoint; the reported Elo values are specific to each protocol. Checkpoints, cohort identities, seeds and locks are pinned by the linked reports and their accompanying protocol and metrics records.

None of these closeouts adopts a checkpoint or establishes learning of independent first proposals, deployment throughput or unseen-family transfer. The final comparisons above and their interpretation in Section 8 remain pending.

8 Discussion

The discussion below is reserved until the pre-registered evaluation in Section 7 is completed and unblinded. **[TBD: Supply a source-grounded discussion of the completed evaluation from Jepson. Reconcile every conclusion below with those outcomes and their uncertainty.]**

8.1 Learning to select and learning to invent

Compare emitted programs against the training archive and behavioral checks to separate retrieval from new composition. **[TBD: Use Jepson's supplied output and archive comparisons, with program behavior checks, to distinguish selection from useful new composition.]**

8.2 Moving judgment across problems

Assess whether task-conditioned program proposals transfer across held-out problem families under matched budgets. **[TBD: Interpret transfer of first-proposal quality across problems using the matched conditions and paired family-level uncertainty supplied by Jepson.]**

8.3 The referee defines the boundary

Examine where proxy validation aligns with or diverges from held-out generalization and where candidates fail. **[TBD: Discuss referee validity, contamination, admission versus generalization, and documented failure limits from Jepson's source.]**

8.4 A bounded claim about self-improvement

Test each level of the self-improvement hierarchy against source-bound parameter updates and held-out proposal outcomes. **[TBD: State which level of self-improvement is supported by source-bound training changes and independent proposal outcomes supplied by Jepson.]**

8.5 Limitations

Document the scope, language, referee and compute boundaries that constrain interpretation of the results. **[TBD: Describe limits on task coverage, the program language, referee reliability and measured cost using Jepson's supplied source.]**

9 Conclusion

Summarize the verified findings and their boundaries once the held-out evaluation record is complete. **[TBD: Supply the conclusion and next steps justified by Jepson's results and their limits; reserve stronger claims until their required evidence is supplied.]**

A Referee Inventory

Section 3 traces the four pillars of Section 2 through the milestones used here: algorithm selection [90], typed genetic programming and evolutionary search [53, 71, 89], self-play and expert iteration [1, 93, 101], and task-conditioned amortization across tabular problems [34, 44, 73]. Each of those systems depends on a referee. Table 9 collects the main text's referees by domain. A domain enters the innovation game only when its referee is reliable and affordable (Section 4.5).

Family allocation is separate from the within-dataset referee. In the native audit path, approved datasets are grouped by source identity, declared family or identical training bytes before seeded family ordering assigns the meta-training, meta-validation and meta-test partitions. This code path does not establish that a particular corpus or checkpoint satisfies the split.

Table 9: **Referees by domain.** Each row restates a referee described in Sections 3 and 4; the last row states the reference referee for algorithm invention.

Domain	Referee	Systems
Board games	The rules of the game and its final score	AlphaGo Zero, AlphaZero [101, 102]
Machine-learning research	A sealed test set that scores every entry	Netflix Prize, ImageNet [7, 29, 92]
Code	Unit tests; a code executor	Codex, Absolute Zero [19, 122]
Matrix multiplication	The rules of TensorGame	AlphaTensor [33]
Sorting	Sorting tests	AlphaDev [66]
Mathematical constructions	Programmatic evaluators	FunSearch [91]
Accelerator kernels	Timers	AlphaEvolve [75]
Formal proofs	The Lean proof checker	AlphaProof, STP [28, 45]
RNA design	Laboratory tests of designs	EteRNA [60]
Algorithm invention (this paper)	Hidden-fold scores against the incumbent; promotion on strict improvement	EurekaZero (Sections 4.4 and 6)

B Incident Log and Audit Safeguards

The reliability requirements in Section 6.6 distinguish a documented baseline exclusion from checks still required for final evaluation. Artifacts that violate held-out isolation belong in the audit record:

- **Excluded test-selected search baseline.** An earlier internal leaderboard entry selected each dataset's best test-scored outcome from 63,108 evaluations in total across 42 datasets. Selection using test-fold scores biases the reported metric upward [17]. The entry is therefore excluded from all comparative claims.
- **Input-closure and canary checks.** Before benchmark release, the evaluation protocol requires passing receipts for an inner-fold label-poison test, a sealed canary check, and a train-only input-closure audit verifying that held-out test targets and their derived statistics do not enter board encoding or candidate generation. The final preregistration and audit receipts belong to the evaluation record in Section 7.1; they are not reproduced here.

C Training Details

These implementation specifications describe the historical reference checkpoint in Section 6, separate from the later adapter snapshot in Section 6.8:

- **Board encoder blocks (Section 6.2).** The frozen 2,048-value context $z_p = \phi_{\mathrm{P}}(p)$ joins three blocks: (i) 512 pooled values from a pretrained row encoder of width 128 across training rows by mean, population standard deviation, and 10th and 90th percentiles; (ii) 1,024 values projected from a 2,048-value third-party tabular regressor state capped at 768 context rows and 256 query rows (the external CPU path first subsamples to at most 1,000 training rows; third-party provenance pending; restricted; redistribution prohibited); and (iii) 512 values from two frozen seed-17 target-summary windows. Each block is normalized before concatenation and clipping to ± 32. Non-finite contexts are refused. Encoding records report wall-clock latency from 11.7 s to 20.8 s across eight reference encoding jobs, within the 30 s cap.
- **Comparison construction (Sections 6.4 and 6.5).** Lists G_p require at least eight valid scored programs per problem, with percentile $\rho_w = 1 - \mathrm{rank}/(|G_p| - 1)$ and writer π_θ sequence weight $0.05 + \rho_w^4$. The native trainer gives problem families equal mass and forms each training comparison from one list: its best program plus one program drawn from each of seven equal rank bands of the rest. Ranking pairs stay within these comparisons; reduction across them is described below. The Python ranking helpers group by dataset identifier. Native ranking groups by corpus context and the eight-row comparison index; separate comparisons from the same context do not pair with each other. The native ranking helper skips non-finite targets and exact ties, $q_w = q_u$, without an exclusion margin or logit rescaling. The trainer refuses non-finite supervision before constructing the loss. Equations (3) and (4) show the list-level form.
- **Sequence supervision (Equation (4)).** The native decoder prepends the projected context token before `BOS`. The sequence loss leaves `BOS` unsupervised but includes the following non-padding tokens, including the engine and regression-task header and `EOS`. That header is fixed during inference. Label smoothing of 0.05 applies over legal tokens; token losses are averaged within each program before the normalized rank-weighted mean over programs. The displayed aggregate coefficients match the native objective; the checkpoint's objective identity and configuration still belong in its evaluation record.
- **Runtime supervision (Equation (5)).** The runtime term compares $\ln(1+s)$ with $\ln(1+\mathrm{softplus}(h))$ by Huber loss with threshold 1, for measured seconds $s \geq 0$ and raw head output h.
- **Ranking targets (Equations (3) and (5)).** In Equation (5), $q_w = 1/(1+\mathrm{RMSE}_w/m_p)$, where m_p is the corpus list's median RMSE. Bradley–Terry weights use $|q_w - q_u|$, doubled when either member is at or above the 90th percentile, then reduced over all non-tied within-comparison pairs in the rank-local microbatch. The implementation mean-scales their weights, casts them to `float32` and averages weighted pair losses. Listwise losses are computed per eligible comparison group and averaged across groups.
- **Registered parameter inventory (Section 6.4).** The network has 27,796,494 parameters in total (six decoder

blocks, width 512, four attention heads, prefix context token, and maximum length 128 tokens over the 401-token grammar). Each of the writer π_θ and ranking-map v_ψ token tables allocates 657 rows: 401 grammar rows plus 256 reserved rows. The reserved rows are never indexed as valid input or output tokens; the tied writer projection is restricted to the grammar rows. The registered total also includes zero-weight stability and loss-scale terms, so it does not measure learned or exercised capacity. Approximately 6.90 million parameters lie outside the writer's token generation path. Decoding bypasses them, but the native checkpoint loader still requires them.

- **Optimization notes (Section 6.5).** The records describe AdamW training (learning rate 10^{-4}, weight decay 0.01, gradient-norm clip 1.0, global batch size 2,048) on eight A100 GPUs in September 2026 over 10,214,332 scored programs from 80 training datasets (7,877 lists; 8,894 contexts). Global batch size alone does not establish per-GPU microbatch or accumulation settings. These historical reference specifications are separate from the later adapter snapshot in Section 6.8. Checkpoint acceptance evidence and a held-out promotion receipt remain required inputs to the evaluation contract (Section 7.1).
- **Execution sandbox and parity (Section 6.6).** The final evaluation protocol requires candidate isolation with `seccomp` and Landlock, `cgroup` memory limits with swap disabled, and native versus emitted-code agreement within $10^{-9}\max(1, |\hat{y}|)$. Isolation and parity receipts from the final evaluator belong to the same evaluation record.

D Flywheel Telemetry

The main text gives four views of adapter decisions and program activity. This appendix adds their timing and event counts (Appendices D.1 and D.2), later operating windows (Appendices D.3 to D.6), and separate adapter-admission and program-promotion rates (Appendices D.7 and D.8). The canonical admission ledger is Table 10; earlier windows and historical GP comparisons are in Appendix D.10. The scope statement in Section 6.8 applies throughout this appendix.

D.1 Development measurements over completed-round time

The third column of Figure 12 shows rounds 2343–2362 alongside the other three 20-round development windows. Total completed-round wall time includes qualification waits, and each point is an end-of-round summary.

D.2 Event counts in retained development windows

Figure 13 summarizes five timed 20-round windows: 2343–2362, 2508–2527, 2644–2663, 2933–2952 and 3001–3020. It pairs per-round event-count distributions with whole-window event rates and mean round durations. The earlier untimed windows are not included in those rates. Figure 12 separately shows trajectories for 2005–2024, 2082–2101, 2343–2362 and 2508–2527; the 2644–2663 trajectory is retained in Figure 7. The windows remain separate under the scope in Section 6.8.

D.3 A subsequent development window

The fourth column of Figure 12 pairs per-round program-event counts with candidate and recorded-best utility for rounds 2508–2527. Its 20 summaries contain 15 strict program-incumbent events and zero adapter admissions. The other columns retain separate earlier windows; Appendix D.10 lists their counts. No gap between captures is filled.

D.4 Utility on its original scale

The bottom panel of the fourth column in Figure 12 shows candidate and recorded-best validation pool utility for rounds 2508–2527 on their original scale. This is the same window as Appendix D.3, with no additional observations. The retained best is 1.1378 (recorded identity: round 882); that identity is not the first round of strict gain.

D.5 Completed-round rates

For the 20 summaries in the fourth column of Figure 12, rounds 2508–2527 total 1,731.45 s and 15 strict program-incumbent events. Dividing total counts by total logged time gives 41.58 completed rounds/h and 31.19 events/h; these are not averages of per-round rates. Figure 13 places the event rate and mean round duration alongside the other timed windows. The underlying per-round times remain in the retained window record; they are not plotted again here.

Time includes logged qualification waits and omits gaps outside completed rounds. None records an adapter admission. Scope: Section 6.8.

D.6 Adapter utility and completed-round throughput

Figure 8 extends the retained operating record with 20 later completed rounds, 2696–2715, captured at 06:54 UTC on 1 November 2026. The summary and admission journals agree on every candidate utility, recorded best, acceptance flag and reported best-round marker in this window. The source retains no adapter admission here.

Throughput counts distinct training contexts within each round and divides by that round's full logged wall time. These within-round counts do not establish the number of distinct tasks across rounds. This rate has a different denominator from training presentations per training second. The separate capture in Figure 15 provides a later operating window under the same completed-round denominator.

D.7 Recorded-best trajectory and cumulative acceptance rate

Figure 14 combines the 1,889 consecutive admission evaluations across rounds 775–2663 with their cumulative acceptance rate and candidate utility-shortfall distribution. The left panel tracks candidate proposals against the post-decision

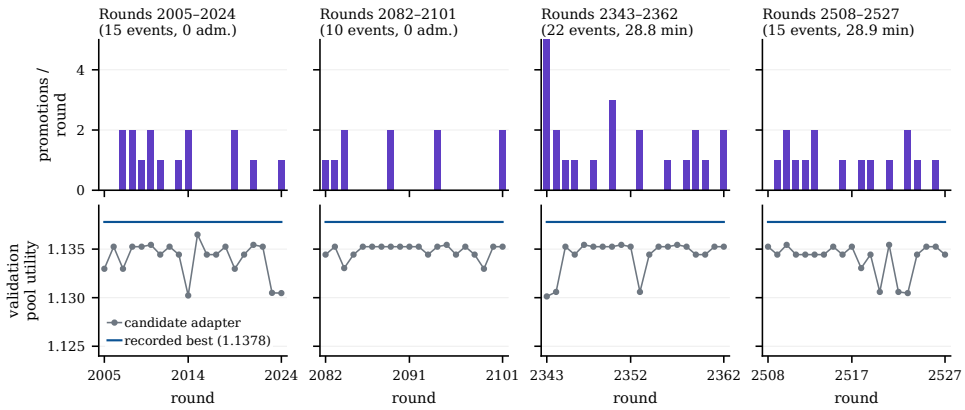


Figure 12: **Training-program promotions and adapter validation across four 20-round windows.** Columns show four separate 20-round summary windows: rounds 2005–2024 (15 events), 2082–2101 (10 events), 2343–2362 (22 events over 28.8 min of completed-round wall time), and 2508–2527 (15 events over 28.9 min). Top row: strict training-program incumbent promotions per completed round. Bottom row: candidate adapter validation pool utility (grey) and the post-decision recorded best (blue, 1.1378 from round 882). Every window records zero adapter admissions. The separate 2644–2663 window is shown once in Figure 7. Scope: Section 6.8.

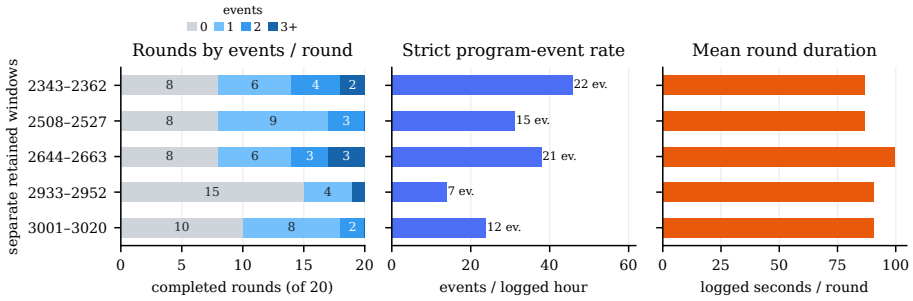


Figure 13: **Event distributions, program-event rates, and round durations across retained 20-round windows.** Each row is a separate capture of 20 consecutive completed rounds with logged wall times. Left: completed rounds grouped by 0, 1, 2, or 3+ strict training-program incumbent events. Center: strict training-program incumbent events divided by total logged round time (events per hour), with total window event counts annotated. Right: mean completed-round duration (seconds) over the 20 rounds, including all within-round work and waits. Windows are displayed separately; program events are distinct from adapter admissions. Scope: Section 6.8.

recorded-best validation pool utility across the two strict improvements (rounds 785 and 859, reaching 1.1376 and 1.1378 from the initial 1.1074) and the two accepted ties (rounds 782 and 882). The center panel tracks the cumulative share of elapsed rounds accepted up to each round: the cumulative rate reaches 18.18% at round 785 ($2/11$ rounds), stands at 3.70% at the last accepted tie ($4/108$ at round 882), and declines across the 1,781 subsequent rejected rounds to 0.21% ($4/1{,}889$) by round 2663.

D.8 Promotions per hour and validation trajectory

Figure 7 retains the trajectory for rounds 2644–2663: per-round and rolling strict training-program promotion rates, with candidate and recorded-best adapter validation utility. Whole-window rates are compared across separate captures in Figure 13.

D.9 Retained admission ledger and pre-decision margins

Figure 6 summarizes complete trailing 50-round windows and cumulative near-tie rejections from the same record, using $\Delta_r = u_{\text{cand},r} - u_{\text{prev},r}$. The default rule accepts $\Delta_r \geq 0$; Table 10

Table 10: **Canonical retained admission ledger.** Every complete row in the captured prefix, without outcome selection. Strict gains and exact ties use unrounded candidate minus prior-best utility. Scope: Section 6.8.

Quantity	Recorded value
Capture cutoff (UTC)	2026-11-01 21:18
Complete rounds	775–3197 (2,423 records)
Admissions / rejections	8 / 2,415
Strict-gain rounds	785, 859, 2750, 3175
Exact-tie rounds	782, 882, 2795, 3100
Final retained utility	`1.1389572403748425`
Recorded best-round identity	3175
New rows since prior capture	176 (2 admitted)
$[-10^{-4}, 0)$ rejected margins	26
$[-10^{-3}, -10^{-4})$ rejected margins	49
Rejected margins below -10^{-3}	2,340
Closest rejected margin	`-0.0000433838086485` (round 803)
Median rejected margin	`-0.0033601045139468`

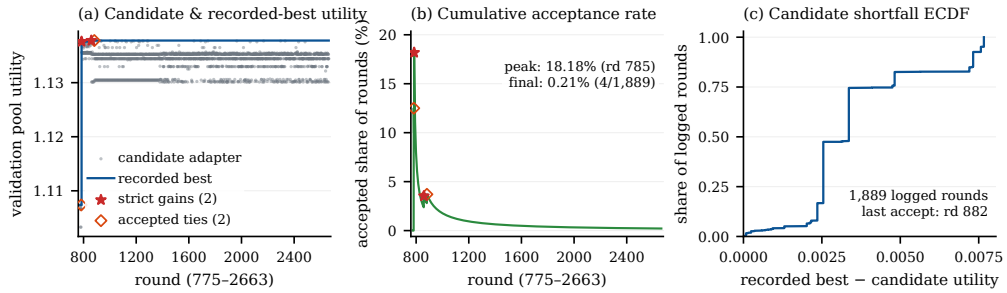


Figure 14: **Adapter validation, cumulative acceptance rate, and candidate utility shortfalls across rounds 775–2663.** All three panels use the same 1,889 consecutive admission evaluations (Table 10). Left: candidate validation utility (logged scalar score; grey) and post-decision recorded best (blue), with stars at strict gains (rounds 785 and 859) and hollow diamonds at accepted ties (rounds 782 and 882). Center: cumulative accepted share of elapsed rounds (%), peaking at 18.18% at round 785 and declining across 1,781 subsequent rejected rounds to 0.21% (4/1,889). Right: empirical cumulative distribution (fraction of evaluations) of recorded-best minus candidate validation utility; zero marks the four accepted evaluations. Scope: Section 6.8.

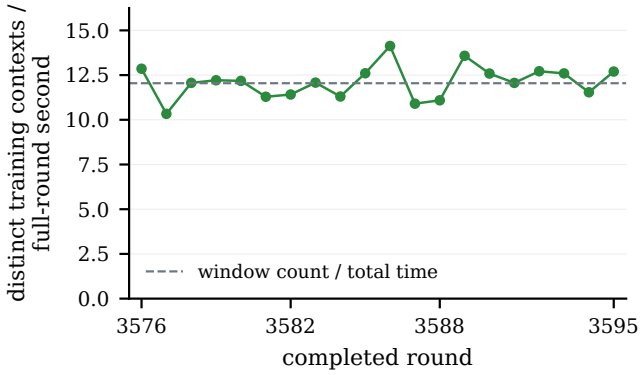


Figure 15: **Completed-round throughput in a fresh live capture.** Twenty consecutive rounds, 3576–3595, captured at 09:54 UTC on 2 November 2026. Each point divides 1,219 distinct training contexts by that round's full wall time (contexts per second), including all work and waits. The dashed line divides the window's summed counts by summed time. Repeated contexts across rounds are not independent tasks; this is an operating observation, not deployment throughput. Scope: Section 6.8.

keeps its chronology, exact counts and margin summaries in one place.

D.10 Earlier admission windows and historical search

The main text selects four views of admission decisions and operating activity. Its 2644–2663 window appears only in Figure 7. The appendix grid pairs the other four retained development windows. Consolidation changes no underlying observation. Figure 14 covers the 775–2663 admission prefix, while Figure 16 resolves the 25 retained detailed summaries and their train–validation aggregates. Gaps between windows are not filled or treated as observations.

Lower predicted cost is preferred. Logged utility averages the selected programs' recorded outcomes; it is not the adapter's cost prediction. For new proposals, the caller uses predicted costs and an exploration setting to select programs for exact evaluation. The referee still decides validity and improvement. Logging an evaluation does not itself add the program to the adapter's training pool.

Figure 14 shows the earlier 775–2663 prefix of the unified ledger in Section 6.8; Appendix D.7 provides its acceptance-rate view. Comparisons across harnesses require matched data, normalization and scoring protocols.

The older detailed summaries cover only 25 of those rounds. In that subset, training utility exceeds validation utility by 0.282–0.290 (Figure 16). This is a difference between reported aggregates on separate training and validation contexts; it does not isolate a cause or establish behavior in the missing summaries. The late validation plateau is evidence to explain, rather than proof of useful transfer. Encoder integrity and parameter updates require their own receipts; the plotted utility tables do not establish either. Figure 16 shows the loop's operating measurements from the earlier retained window.

Retained operating windows. In the five earlier 20-round program-event slices, rounds 2005–2024 record 15 events (Figure 12), rounds 2082–2101 record 10 (Figure 12), rounds 2343–2362 record 22 (Figure 12), rounds 2508–2527 record 15 (Figure 12), and rounds 2644–2663 record 21 (Figure 7). Each slice records zero adapter admissions. The later 20-round throughput slice, 2696–2715, also records none; it reports contexts and wall time rather than a program-event total. Completed-time, event-distribution and rate summaries appear in Appendices D.1, D.2, D.5, D.6 and D.8. The window separation and scope follow Section 6.8; logged round time includes work and qualification waits.

The right panel of Figure 14 summarizes candidate gaps to each round's recorded best. Each retained round contributes once; the curve does not count independent tasks or distinct programs.

Historical GP target scores provide a separate view in the first panel of Figure 17. The 159 repeated dataset/cut/arm groups cover 55 distinct dataset/cut pairs; the 64 groups with lower final scores cover 23 pairs. Shared incumbent and teacher seeds make the arms dependent. Scope: Section 6.8.

The remaining three panels of Figure 17 follow one historical dataset/cut, chosen after inspection for the largest policy-seeded relative improvement in the preceding comparison set. Every reported generation is shown, without applying

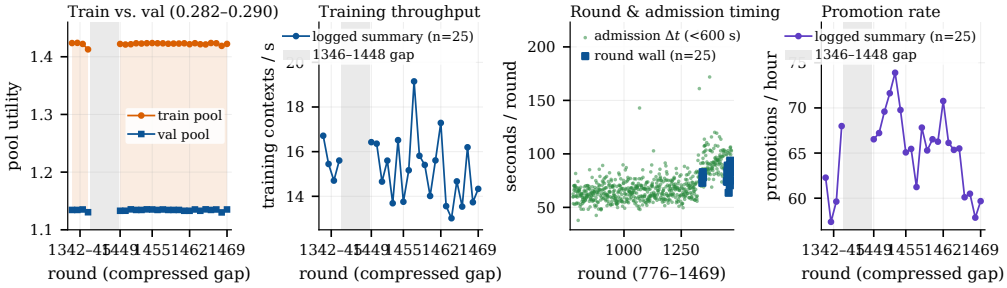


Figure 16: **Train–validation aggregates and operating measurements from earlier retained windows.** First, second, and fourth panels: 25 retained summaries spanning rounds 1342–1345 and 1449–1469 (rounds 1346–1448 were not retained), showing train versus validation pool utility (gap 0.282–0.290), distinct training contexts per completed-round second, and strict training-program promotions per hour over up to 20 completed rounds including qualification waits. Third panel: consecutive admission-timestamp intervals: all 695 admission rows (775–1469) carry timestamps, giving 694 consecutive pairs. The panel plots the 692 positive differences below 600 s (green), excluding two longer pauses, alongside the 25 measured round wall times (blue squares). Scope: Section 6.8.

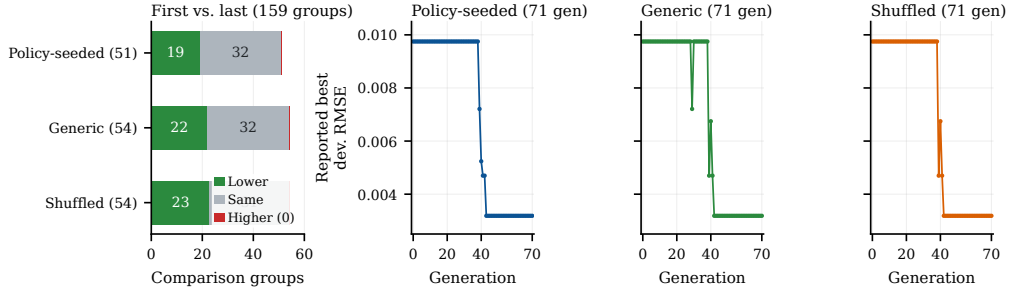


Figure 17: **Historical GP target progress and post-hoc 71-generation trajectories.** First panel: first-to-last reported best development RMSE across all 159 repeated comparison groups (55 distinct dataset/cut pairs) at 1,000 evaluations per search, grouped by search arm. Second through fourth panels: all 71 generations per arm for one historical dataset/cut selected post hoc for the largest policy-seeded relative improvement. RMSE is in historical evaluator units (lower is better). The arms shared incumbent and teacher seeds, and elapsed time varied; these development records do not establish independent discoveries or a learned-policy advantage.

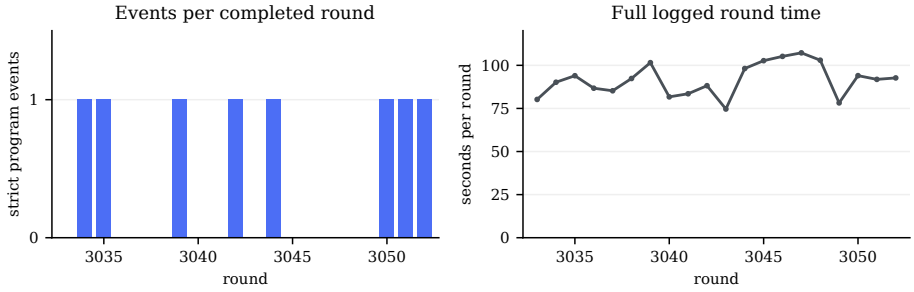


Figure 18: **Program-event yield and full round time in a later capture.** Every completed summary in rounds 3033–3052, including zero-event rounds. Left: strict training-program incumbent events per completed round (count). Right: each round's full logged wall time (seconds), including work and waits. The window records 8 events over 1,831.32 s (15.73 events per logged hour) and no adapter admissions. These are distinct event and time measures. Scope: Section 6.8.

a cumulative minimum. All three arms finish at the same value; the policy-seeded arm first records it one generation after the controls. Shared seeds prevent treating their matching outcomes as independent discoveries.

D.11 Admission frequency through a later cutoff

Figure 19 groups the complete admission journal from the earlier ledger's first round (Table 10) through the latest complete admission matched to a retained summary. Its 24 blocks contain 23 full blocks and one partial block; 19 have no admission, including 17 consecutive full blocks in rounds 975–2674. The 21 available summaries (3145–3165) match the journal's scores and decisions; older rows are journal-only. The last accepted decision, at round 3100, is an exact tie, followed by 65 rejections. Frequencies count decisions, not independent tasks, distinct programs or a stationary admission probability. Stable log-prefix hashes bind this separate capture. The earlier ledger and trace data remain unchanged; the field-binding limits in Section 6.8 still apply.

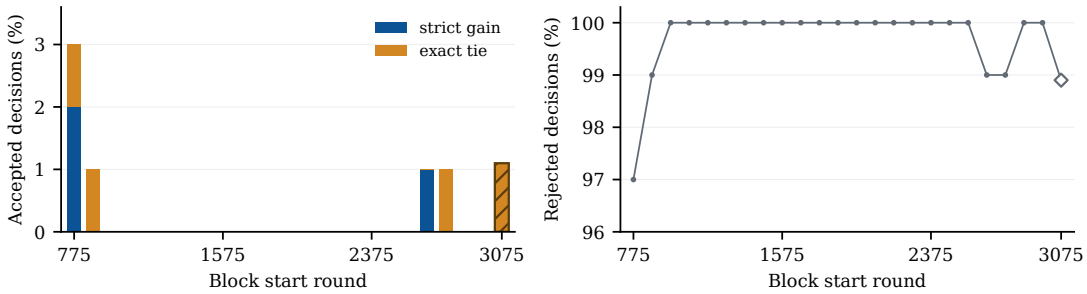


Figure 19: **Adapter admission and rejection frequencies by observed-round block.** All 2,391 complete journal decisions in rounds 775–3165, captured at 20:17 UTC on 1 November 2026. Blocks contain 100 consecutive observed rounds; the final 3075–3165 block contains 91. Left: accepted strict gains (blue) and exact ties (orange), stacked percentages. Right: rejected decisions (percent; axis starts at 96%). Each denominator is that block's observed rounds; hatching and a diamond mark the partial block. Totals: 3 strict gains, 4 exact ties and 2,384 rejections. These are adaptive adapter decisions, distinct from program-incumbent events and independent first-proposal performance. Scope: Section 6.8.

D.12 Program-event rate and round duration across windows

Figure 13 consolidates four retained windows with raw event counts and wall durations and a fresh matched 20-round capture. Each rate divides that window's total events by its total completed-round wall time; each mean duration divides the same time by 20. Sources lacking raw duration, or lacking 20 consecutive rounds, are excluded rather than imputed. The last of those five windows, rounds 3001–3020, records 12 program events over 1,809.50 s and no adapter admissions. Its summary event counts also match the separate promotion journal.

These are separate adaptive training-search windows, with the source and evaluation binding limits in Section 6.8. Their ordering does not establish a learning curve, an independent policy comparison or serving throughput.

D.13 A later program-event window

Figure 18 pairs event counts with durations rather than treating every round as equal time. This separate capture was taken at 16:51 UTC on 1 November 2026; all 20 summary event counts match the promotion journal and the summaries match their admission records. It does not extend the earlier admission prefix in Table 10 or join the gaps between retained windows. Pricing a proposal still needs the attempt ledger in Section 5.6.

D.14 Candidate and retained-best decisions in a recent capture

Figure 20 shows the latest 500 consecutive complete admission records matched to the separate per-round audit at capture. This window is separate from the earlier retained views; no intervening gaps are joined. The scope and unresolved source-binding requirements of Section 6.8 still apply.

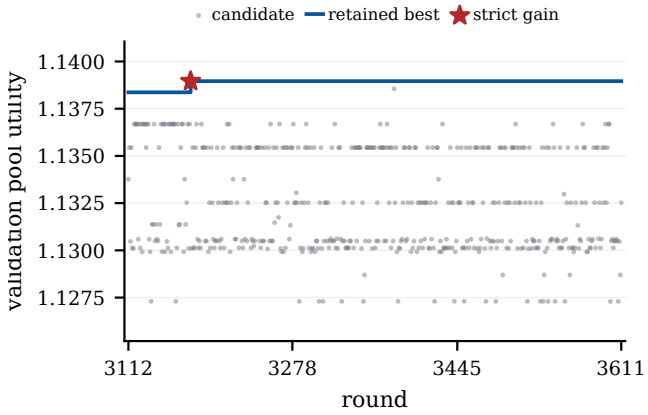


Figure 20: **Recent candidate and retained-best adapter trajectory.** Rounds 3112–3611, captured at 10:25 UTC on 2 November 2026. Grey points show candidate utility; blue is the post-decision retained best (logged dimensionless utility, larger preferred). The star marks the one accepted strict gain, at round 3175; the other 499 decisions are rejections. Classification uses exact logged values, with no rounding margin. These are adaptive decisions on repeatedly used development contexts, not independent first-proposal results or evidence of writer/value-net lift. Scope: Section 6.8.

E A Later Program-Incumbent Event Clock

Figure 21 uses a separate capture of 20 consecutive completed summaries, with every program-event count checked against the promotion journal. The full-round clock includes all work and waits, including rounds with no event. These records retain the adaptive training-side scope and unresolved source-binding requirements of Section 6.8; independent first-proposal outcomes remain pending.

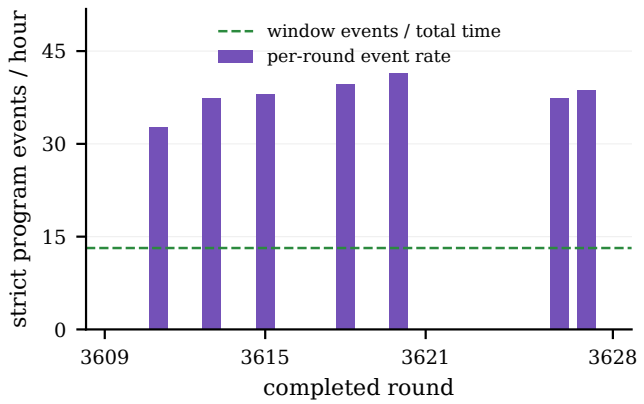


Figure 21: **Strict training-program incumbent events per logged hour.** Rounds 3609–3628, captured at 10:56 UTC on 2 November 2026. Bars divide each round's event count by its full wall time (events per hour); the dashed line divides the seven events by the total 1,914.91 s (13.16 events per hour). All 20 rounds are retained, including 13 with no event. The window records zero adapter admissions; these program events do not measure independent first-proposal or serving performance. Scope: Section 6.8.

F Sizes of Strict Program-Incumbent Gains

Figure 22 adds event magnitudes to the counts and clocks above. Each observation is a journaled strict training-program incumbent change, not an adapter admission. The distribution conditions on an event; rounds without one are counted separately. Logged scalar increments are not Elo changes or independent performance estimates.

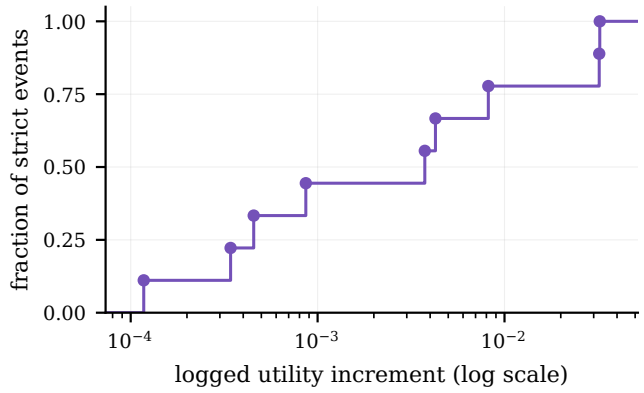


Figure 22: **Distribution of logged strict training-program utility increments.** Rounds 3656–3675, captured at 12:25 UTC on 2 November 2026. The empirical cumulative distribution gives each of the nine strict events equal weight; the horizontal axis is logarithmic. An increment is the journaled new utility minus its previous incumbent utility, computed from unrounded values. All 20 completed rounds were checked against the journal; 13 had no event and contribute no zero-gain observation. The window records zero adapter admissions. Event-level confidence intervals are unavailable; the curve is descriptive, without uncertainty bands. Scope: Section 6.8.

References

[1] T. Anthony, Z. Tian, and D. Barber. Thinking fast and slow with deep learning and tree search. In *Advances in Neural Information Processing Systems*, volume 30, pages 5360–5370, 2017. doi: 10.48550/arXiv.1705.08439. URL https://proceedings.neurips.cc/paper_files/paper/2017/file/d8e1344e27a5b08cdfd5d027d9b8d6de-Paper.pdf.

[2] P. Azoulay, C. Fons-Rosen, and J. S. Graff Zivin. Does science advance one funeral at a time? *American Economic Review*, 109(8): 2889–2920, 2019. doi: 10.1257/aer.20161574.

[3] L. Bailey, K. Wen, K. Dong, T. Hashimoto, and T. Ma. Scaling self-play with self-guidance. arXiv:2604.20209, 2026. URL https://arxiv.org/abs/2604.20209.

[4] D. Balduzzi, K. Tuyls, J. Pérolat, and T. Graepel. Re-evaluating evaluation. In *Advances in Neural Information Processing Systems*, volume 31, pages 3268–3279, 2018. doi: 10.48550/arXiv.1806.02643. URL https://arxiv.org/abs/1806.02643.

[5] M. Balog, A. L. Gaunt, M. Brockschmidt, S. Nowozin, and D. Tarlow. DeepCoder: Learning to write programs. In *International Conference on Learning Representations (ICLR)*, 2017. doi: 10.48550/arXiv.1611.01989. URL https://arxiv.org/abs/1611.01989.

[6] R. Bellman. *Dynamic Programming*. Princeton University Press, Princeton, NJ, 1957. doi: 10.1515/9781400835386.

[7] J. Bennett and S. Lanning. The Netflix Prize. In *Proceedings of the KDD Cup and Workshop 2007*, 2007. URL https://www.cs.uic.edu/~liub/KDD-cup-2007/proceedings/The-Netflix-Prize-Bennett.pdf.

[8] N. Bloom, C. I. Jones, J. Van Reenen, and M. Webb. Are ideas getting harder to find? *American Economic Review*, 110(4):1104–1144, 2020. doi: 10.1257/aer.20180338.

[9] M. A. Boden. *The Creative Mind: Myths and Mechanisms*. Weidenfeld & Nicolson, London, 1990. doi: 10.4324/9780203508527.

[10] M. A. Boden. Creativity and artificial intelligence. *Artificial Intelligence*, 103(1–2):347–356, 1998. doi: 10.1016/S0004-3702(98)00055-1.

[11] J. L. Borges. La biblioteca de Babel. In *El jardín de senderos que se bifurcan*. Sur, Buenos Aires, 1941. Printed 30 December 1941, issued 1942 (LCCN 43002084). English translation by James E. Irby, "The Library of Babel", in *Labyrinths*, New Directions, 1962.

[12] R. A. Bradley and M. E. Terry. Rank analysis of incomplete block designs: I. The method of paired comparisons. *Biometrika*, 39(3/4): 324–345, 1952. doi: 10.1093/biomet/39.3-4.324.

[13] R. Brandom. What happens when AI starts building itself? TechCrunch, 14 May 2026. URL https://techcrunch.com/2026/05/14/what-happens-when-ai-starts-building-itself/.

[14] T. Brooks, B. Peebles, C. Holmes, W. DePue, Y. Guo, L. Jing, D. Schnurr, J. Taylor, T. Luhman, E. Luhman, C. Ng, R. Wang, and A. Ramesh. Video generation models as world simulators. OpenAI technical report, February 2024. URL https://openai.com/index/video-generation-models-as-world-simulators/.

[15] D. T. Campbell. Blind variation and selective retention in creative thought as in other knowledge processes. *Psychological Review*, 67 (6):380–400, 1960. doi: 10.1037/h0040373.

[16] Z. Cao, T. Qin, T.-Y. Liu, M.-F. Tsai, and H. Li. Learning to rank: From pairwise approach to listwise approach. In *Proceedings of the 24th International Conference on Machine Learning (ICML)*, pages 129–136, 2007. doi: 10.1145/1273496.1273513.

[17] G. C. Cawley and N. L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11(70):2079–2107, 2010. URL https://www.jmlr.org/papers/v11/cawley10a.html.

[18] W. G. Chase and H. A. Simon. Perception in chess. *Cognitive Psychology*, 4(1):55–81, 1973. doi: 10.1016/0010-0285(73)90004-2.

[19] M. Chen, J. Tworek, H. Jun, Q. Yuan, H. P. d. O. Pinto, J. Kaplan, H. Edwards, Y. Burda, N. Joseph, G. Brockman, A. Ray, R. Puri, G. Krueger, M. Petrov, H. Khlaaf, G. Sastry, P. Mishkin, B. Chan, S. Gray, N. Ryder, M. Pavlov, A. Power, Ł. Kaiser, M. Bavarian, C. Winter, P. Tillet, F. P. Such, D. W. Cummings, M. Plappert,

F. Chantzis, E. Barnes, A. Herbert-Voss, W. H. Guss, A. Nichol, A. Paino, N. A. Tezak, J. Tang, I. Babuschkin, S. Balaji, S. Jain, W. Saunders, C. Hesse, A. N. Carr, J. Leike, J. Achiam, V. Misra, E. Morikawa, A. Radford, M. M. Knight, M. Brundage, M. Murati, K. Mayer, P. Welinder, B. McGrew, D. Amodei, S. McCandlish, I. Sutskever, and W. Zaremba. Evaluating large language models trained on code. arXiv:2107.03374, 2021. URL https://arxiv.org/abs/2107.03374.

[20] X. Chen, C. Liang, D. Huang, E. Real, K. Wang, H. Pham, X. Dong, T. Luong, C.-J. Hsieh, Y. Lu, and Q. V. Le. Symbolic discovery of optimization algorithms. In *Advances in Neural Information Processing Systems*, volume 36, pages 49205–49233, 2023. doi: 10.48550/arXiv.2302.06675. URL https://arxiv.org/abs/2302.06675.

[21] Z. Chen, Y. Deng, H. Yuan, K. Ji, and Q. Gu. Self-play fine-tuning converts weak language models to strong language models. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 6621–6642, 2024. doi: 10.48550/arXiv.2401.01335. URL https://arxiv.org/abs/2401.01335.

[22] P. F. Christiano, J. Leike, T. B. Brown, M. Martic, S. Legg, and D. Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*, volume 30, pages 4299–4307, 2017. doi: 10.48550/arXiv.1706.03741. URL https://arxiv.org/abs/1706.03741.

[23] S. Cooper, F. Khatib, A. Treuille, J. Barbero, J. Lee, M. Beenen, A. Leaver-Fay, D. Baker, Z. Popović, and Foldit Players. Predicting protein structures with a multiplayer online game. *Nature*, 466(7307): 756–760, 2010. doi: 10.1038/nature09304.

[24] H. Dai, Y. Tian, B. Dai, S. Skiena, and L. Song. Syntax-directed variational autoencoder for structured data. In *International Conference on Learning Representations (ICLR)*, 2018. doi: 10.48550/arXiv.1802.08786. URL https://arxiv.org/abs/1802.08786.

[25] C. Darwin. *On the Origin of Species by Means of Natural Selection, or the Preservation of Favoured Races in the Struggle for Life*. John Murray, London, 1859. doi: 10.5962/bhl.title.68064. URL http://darwin-online.org.uk/content/frameset?itemID=F373&viewtype=text&pageseq=1.

[26] A. D. de Groot. *Thought and Choice in Chess*. Mouton, The Hague, 1965. doi: 10.5117/9789053569986.

[27] DeepSeek-AI, A. Liu, B. Feng, B. Xue, B. Wang, B. Wu, C. Lu, C. Zhao, C. Deng, C. Zhang, C. Ruan, D. Dai, D. Guo, D. Yang, D. Chen, et al. DeepSeek-V3 technical report. arXiv:2412.19437, 2024. URL https://arxiv.org/abs/2412.19437.

[28] K. Dong and T. Ma. STP: Self-play LLM theorem provers with iterative conjecturing and proving. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pages 14114–14136, 2025. doi: 10.48550/arXiv.2502.00212. URL https://proceedings.mlr.press/v267/dong25h.html.

[29] D. Donoho. 50 years of data science. *Journal of Computational and Graphical Statistics*, 26(4):745–766, 2017. doi: 10.1080/10618600.2017.1384734.

[30] S. Du, X. Yan, J. Shi, Z. Cao, S. Feng, Z. Liang, B. Sun, T. Peng, Y. Zhou, X. Li, J. Zhou, L. He, B. Zhang, and L. Bai. MLEvolve: A self-evolving framework for automated machine learning algorithm discovery. arXiv:2606.06473, 2026. URL https://arxiv.org/abs/2606.06473.

[31] K. Ellis, C. Wong, M. Nye, M. Sablé-Meyer, L. Morales, L. Hewitt, L. Cary, A. Solar-Lezama, and J. B. Tenenbaum. DreamCoder: Bootstrapping inductive program synthesis with wake-sleep library learning. In *Proceedings of the 42nd ACM SIGPLAN International Conference on Programming Language Design and Implementation (PLDI)*, pages 835–850, 2021. doi: 10.1145/3453483.3454080.

[32] A. E. Elo. *The Rating of Chessplayers, Past and Present*. Arco, New York, 1978.

[33] A. Fawzi, M. Balog, A. Huang, T. Hubert, B. Romera-Paredes, M. Barekatain, A. Novikov, F. J. R. Ruiz, J. Schrittwieser, G. Swirszcz, D. Silver, D. Hassabis, and P. Kohli. Discovering faster matrix multiplication algorithms with reinforcement learning. *Nature*, 610(7930):47–53, 2022. doi: 10.1038/s41586-022-05172-4.

[34] M. Fey, V. Kocijan, F. Lopez, J. E. Lenssen, and J. Leskovec. KumoRFM: A relational foundation model. Kumo Technical Report, May 2025. URL https://docs.nvidia.com/sdgm/research/concepts-models/kumorfm-paper.

[35] R. M. Friedberg. A learning machine: Part I. *IBM Journal of Research and Development*, 2(1):2–13, 1958. doi: 10.1147/rd.21.0002.

[36] I. J. Good. Speculations concerning the first ultraintelligent machine. In F. L. Alt and M. Rubinoff, editors, *Advances in Computers*, volume 6, pages 31–88. Academic Press, 1966. doi: 10.1016/S0065-2458(08)60418-0.

[37] J. Gottweis, W.-H. Weng, A. Daryin, T. Tu, P. Sirkovic, A. Myaskovsky, G. Glowaty, F. Weissenberger, A. Orlandi, D. Popovici, A. Palepu, K. Rong, R. Tanno, K. Saab, F. Zhang, J. Blum, A. Carroll, K. Kulkarni, N. Tomasev, D. Zverinski, I. Rendulic, E. Vedadi, F. Hasler, L. Rimanic, M. Boia, I. Budiselic, B. Feinstein, M. Bellaiche, T. Sheffer, J. Freyberg, J. Ratcliff, O. Bertolli, K. Chou, A. Hassidim, B. Gokturk, A. Vahdat, Y. Guan, V. Dhillon, E. D. Vaishnav, B. Lee, T. R. D. Costa, J. R. Penadés, G. Peltz, Y. Matias, J. Manyika, D. Hassabis, Y. Xu, P. Kohli, A. Pawlosky, A. Karthikesalingam, and V. Natarajan. Accelerating scientific discovery with Co-Scientist. *Nature*, 655(8122):487–496, 2026. doi: 10.1038/s41586-026-10644-y. Preprint: arXiv:2502.18864 (2025), "Towards an AI co-scientist".

[38] S. Gulwani. Automating string processing in spreadsheets using input-output examples. In *Proceedings of the 38th ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages (POPL)*, pages 317–330, 2011. doi: 10.1145/1926385.1926423.

[39] D. Guo, D. Yang, H. Zhang, J. Song, R. Zhang, R. Xu, Q. Zhu, S. Ma, P. Wang, X. Bi, X. Zhang, X. Yu, Y. Wu, Z. F. Wu, Z. Gou, et al. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature*, 645(8081):633–638, 2025. doi: 10.1038/s41586-025-09422-z.

[40] R. W. Hamming. You and your research. Bell Communications Research Colloquium Seminar, Morris Research and Engineering Center, Morristown, NJ, 1986. URL https://hdl.handle.net/10945/37504. Talk given 7 March 1986; transcribed by J. F. Kaiser.

[41] D. Hassabis and A. Kantrowitz. Google DeepMind CEO Demis Hassabis: The path to AGI, LLM creativity, and Google smart glasses. Big Technology Podcast, 23 January 2025. URL https://www.bigtechnology.com/p/google-deepmind-ceo-demis-hassabis.

[42] J. Hoffmann, S. Borgeaud, A. Mensch, E. Buchatskaya, T. Cai, E. Rutherford, D. de Las Casas, L. A. Hendricks, J. Welbl, A. Clark, T. Hennigan, E. Noland, K. Millican, G. van den Driessche, B. Damoc, A. Guy, S. Osindero, K. Simonyan, E. Elsen, J. W. Rae, O. Vinyals, and L. Sifre. Training compute-optimal large language models. In *Advances in Neural Information Processing Systems*, volume 35, pages 30016–30030, 2022. doi: 10.48550/arXiv.2203.15556. URL https://arxiv.org/abs/2203.15556.

[43] J. H. Holland. *Adaptation in Natural and Artificial Systems*. University of Michigan Press, Ann Arbor, MI, 1975. doi: 10.7551/mitpress/1090.001.0001.

[44] N. Hollmann, S. Müller, L. Purucker, A. Krishnakumar, M. Körfer, S. B. Hoo, R. T. Schirrmeister, and F. Hutter. Accurate predictions on small data with a tabular foundation model. *Nature*, 637(8045): 319–326, 2025. doi: 10.1038/s41586-024-08328-6.

[45] T. Hubert, R. Mehta, L. Sartran, M. Z. Horváth, G. Žužić, E. Wieser, A. Huang, J. Schrittwieser, Y. Schroecker, H. Masoom, O. Bertolli, T. Zahavy, A. Mandhane, J. Yung, I. Beloshapka, B. Ibarz, V. Veeriah, L. Yu, O. Nash, P. Lezeau, S. Mercuri, C. Sönne, B. Mehta, A. Davies, D. Zheng, F. Pedregosa, Y. Li, I. von Glehn, M. Rowland, S. Albanie, A. Velingker, S. Schmitt, E. Lockhart, E. Hughes, H. Michalewski, N. Sonnerat, D. Hassabis, P. Kohli, and D. Silver. Olympiad-level formal mathematical reasoning with reinforcement learning. *Nature*, 651(8106):607–613, 2026. doi: 10.1038/s41586-025-09833-y. Published online 12 November 2025.

[46] B. F. Jones. The burden of knowledge and the "Death of the Renaissance man": Is innovation getting harder? *Review of Economic*

Studies, 76(1):283–317, 2009. doi: 10.1111/j.1467-937X.2008.00531.x.

[47] J. Jumper, R. Evans, A. Pritzel, T. Green, M. Figurnov, O. Ronneberger, K. Tunyasuvunakool, R. Bates, A. Žídek, A. Potapenko, A. Bridgland, C. Meyer, S. A. A. Kohl, A. J. Ballard, A. Cowie, B. Romera-Paredes, S. Nikolov, R. Jain, J. Adler, T. Back, S. Petersen, D. Reiman, E. Clancy, M. Zielinski, M. Steinegger, M. Pacholska, T. Berghammer, S. Bodenstein, D. Silver, O. Vinyals, A. W. Senior, K. Kavukcuoglu, P. Kohli, and D. Hassabis. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596 (7873):583–589, 2021. doi: 10.1038/s41586-021-03819-2.

[48] J. Kaplan, S. McCandlish, T. Henighan, T. B. Brown, B. Chess, R. Child, S. Gray, A. Radford, J. Wu, and D. Amodei. Scaling laws for neural language models. arXiv:2001.08361, 2020. URL https://arxiv.org/abs/2001.08361.

[49] A. Karpathy. `autoresearch`: AI agents running research on single-GPU nanochat training automatically. GitHub repository, 6 March 2026. URL https://github.com/karpathy/autoresearch.

[50] F. Khatib, F. DiMaio, Foldit Contenders Group, Foldit Void Crushers Group, S. Cooper, M. Kazmierczyk, M. Gilski, S. Krzywda, H. Zábranská, I. Pichová, J. Thompson, Z. Popović, M. Jaskolski, and D. Baker. Crystal structure of a monomeric retroviral protease solved by protein folding game players. *Nature Structural & Molecular Biology*, 18(10):1175–1177, 2011. doi: 10.1038/nsmb.2119.

[51] Kimi Team. Kimi K2: Open agentic intelligence. arXiv:2507.20534, 2025. URL https://arxiv.org/abs/2507.20534.

[52] A. Koestler. *The Act of Creation*. Hutchinson, London, 1964.

[53] J. R. Koza. *Genetic Programming: On the Programming of Computers by Means of Natural Selection*. MIT Press, Cambridge, MA, 1992. URL https://mitpress.mit.edu/9780262111706/genetic-programming/.

[54] A. Krizhevsky, I. Sutskever, and G. E. Hinton. ImageNet classification with deep convolutional neural networks. In *Advances in Neural Information Processing Systems*, volume 25, pages 1097–1105, 2012. doi: 10.1145/3065386. URL https://papers.nips.cc/paper/4824-imagenet-classification-with-deep-convolutional-neural-networks. Republished in *Communications of the ACM*, 60(6):84–90, 2017.

[55] T. S. Kuhn. *The Structure of Scientific Revolutions*. University of Chicago Press, Chicago, 1962. doi: 10.7208/chicago/9780226458144.001.0001.

[56] M. J. Kusner, B. Paige, and J. M. Hernández-Lobato. Grammar variational autoencoder. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, volume 70 of *Proceedings of Machine Learning Research*, pages 1945–1954, 2017. doi: 10.48550/arXiv.1703.01925. URL https://proceedings.mlr.press/v70/kusner17a.html.

[57] T. Lattimore and C. Szepesvári. *Bandit Algorithms*. Cambridge University Press, Cambridge, 2020. doi: 10.1017/9781108571401.

[58] Y. LeCun and A. Kantrowitz. Why can't AI make its own discoveries? — With Yann LeCun. Big Technology Podcast, 19 March 2025. URL https://podscripts.co/podcasts/big-technology-podcast/why-cant-ai-make-its-own-discoveries-with-yann-lecun.

[59] Y. LeCun, Y. Bengio, and G. Hinton. Deep learning. *Nature*, 521 (7553):436–444, 2015. doi: 10.1038/nature14539.

[60] J. Lee, W. Kladwang, M. Lee, D. Cantu, M. Azizyan, H. Kim, A. Limpaecher, S. Yoon, A. Treuille, R. Das, and EteRNA Participants. RNA design rules from a massive open laboratory. *Proceedings of the National Academy of Sciences*, 111(6):2122–2127, 2014. doi: 10.1073/pnas.1313039111.

[61] Y.-J. Lee, S. Kim, M. Kang, A. C. L. Chuen, Z. Chen, S. Han, T. Jung, and D. Kang. Evolution Fine-Tuning: Learning to discover across 371 optimization tasks. arXiv:2606.29082, 2026. URL https://arxiv.org/abs/2606.29082.

[62] J. Lehman and K. O. Stanley. Abandoning objectives: Evolution through the search for novelty alone. *Evolutionary Computation*, 19 (2):189–223, 2011. doi: 10.1162/EVCO_a_00025.

[63] D. B. Lenat. *AM: An Artificial Intelligence Approach to Discovery in Mathematics as Heuristic Search*. PhD thesis, Stanford University, 1976. Stanford AI Lab Memo AIM-286; report STAN-CS-76-570.

[64] D. B. Lenat. EURISKO: A program that learns new heuristics and domain concepts. The nature of heuristics III: Program design and results. *Artificial Intelligence*, 21(1–2):61–98, 1983. doi: 10.1016/S0004-3702(83)80005-8.

[65] C. Lu, C. Lu, R. T. Lange, Y. Yamada, S. Hu, J. Foerster, D. Ha, and J. Clune. Towards end-to-end automation of AI research. *Nature*, 651 (8107):914–919, 2026. doi: 10.1038/s41586-026-10265-5. Preprint: arXiv:2408.06292 (2024), "The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery".

[66] D. J. Mankowitz, A. Michi, A. Zhernov, M. Gelmi, M. Selvi, C. Paduraru, E. Leurent, S. Iqbal, J.-B. Lespiau, A. Ahern, T. Köppe, K. Millikin, S. Gaffney, S. Elster, J. Broshear, C. Gamble, K. Milan, R. Tung, M. Hwang, T. Cemgil, M. Barekatain, Y. Li, A. Mandhane, T. Hubert, J. Schrittwieser, D. Hassabis, P. Kohli, M. Riedmiller, O. Vinyals, and D. Silver. Faster sorting algorithms discovered using deep reinforcement learning. *Nature*, 618(7964):257–263, 2023. doi: 10.1038/s41586-023-06004-9.

[67] H. Massalin. Superoptimizer: A look at the smallest program. In *Proceedings of the 2nd International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS II)*, pages 122–126, 1987. doi: 10.1145/36206.36194.

[68] A. Merchant, S. Batzner, S. S. Schoenholz, M. Aykol, G. Cheon, and E. D. Cubuk. Scaling deep learning for materials discovery. *Nature*, 624(7990):80–85, 2023. doi: 10.1038/s41586-023-06735-9.

[69] C. Metz. In two moves, AlphaGo and Lee Sedol redefined the future. Wired, 16 March 2016. URL https://www.wired.com/2016/03/two-moves-alphago-lee-sedol-redefined-future/.

[70] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, S. Petersen, C. Beattie, A. Sadik, I. Antonoglou, H. King, D. Kumaran, D. Wierstra, S. Legg, and D. Hassabis. Human-level control through deep reinforcement learning. *Nature*, 518(7540): 529–533, 2015. doi: 10.1038/nature14236.

[71] D. J. Montana. Strongly typed genetic programming. *Evolutionary Computation*, 3(2):199–230, 1995. doi: 10.1162/evco.1995.3.2.199.

[72] J.-B. Mouret and J. Clune. Illuminating search spaces by mapping elites. arXiv:1504.04909, 2015. URL https://arxiv.org/abs/1504.04909.

[73] A. Müller, C. Curino, and R. Ramakrishnan. MotherNet: Fast training and inference via hyper-network transformers. In *International Conference on Learning Representations (ICLR)*, 2025. doi: 10.48550/arXiv.2312.08598. URL https://arxiv.org/abs/2312.08598.

[74] J. Nielsen and J. D. Keasling. Engineering cellular metabolism. *Cell*, 164(6):1185–1197, 2016. doi: 10.1016/j.cell.2016.02.004.

[75] A. Novikov, N. Vũ, M. Eisenberger, E. Dupont, P.-S. Huang, A. Z. Wagner, S. Shirobokov, B. Kozlovskii, F. J. R. Ruiz, A. Mehrabian, M. P. Kumar, A. See, S. Chaudhuri, G. Holland, A. Davies, S. Nowozin, P. Kohli, and M. Balog. AlphaEvolve: A coding agent for scientific and algorithmic discovery. arXiv:2506.13131, 2025. URL https://arxiv.org/abs/2506.13131.

[76] NVIDIA. NVIDIA H100 Tensor Core GPU: Specifications. NVIDIA Data Center Documentation, 2026. URL https://www.nvidia.com/en-us/data-center/h100/.

[77] J. Oh, G. Farquhar, I. Kemaev, D. A. Calian, M. Hessel, L. Zintgraf, S. Singh, H. van Hasselt, and D. Silver. Discovering state-of-the-art reinforcement learning algorithms. *Nature*, 648(8093):312–319, 2025. doi: 10.1038/s41586-025-09761-x.

[78] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. L. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, J. Schulman, J. Hilton, F. Kelton, L. Miller, M. Simens, A. Askell, P. Welinder, P. Christiano, J. Leike, and R. Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744, 2022. doi: 10.48550/arXiv.2203.02155. URL https://arxiv.org/abs/2203.02155.

[79] M. Park, E. Leahey, and R. J. Funk. Papers and patents are becoming less disruptive over time. *Nature*, 613(7942):138–144, 2023. doi: 10.1038/s41586-022-05543-x.

[80] D. Patel and D. Amodei. Dario Amodei (Anthropic CEO) — The hidden pattern behind every AI breakthrough. Dwarkesh Podcast, 8 August 2023. URL https://www.dwarkesh.com/p/dario-amodei.

[81] C. S. Peirce. *Collected Papers of Charles Sanders Peirce, Volume V: Pragmatism and Pragmaticism*. Harvard University Press, Cambridge, MA, 1934. Edited by Charles Hartshorne and Paul Weiss.

[82] B. Pfahringer, H. Bensusan, and C. Giraud-Carrier. Meta-learning by landmarking various learning algorithms. In *Proceedings of the 17th International Conference on Machine Learning (ICML)*, pages 743–750. Morgan Kaufmann, 2000. URL https://ml.cms.waikato.ac.nz/publications/2000/landmarking_icml2000.pdf.

[83] H. Poincaré. Mathematical creation. *The Monist*, 20(3):321–335, 1910. doi: 10.1093/monist/20.3.321. Translated by George Bruce Halsted.

[84] M. Polanyi. *The Tacit Dimension*. Doubleday, Garden City, NY, 1966.

[85] G. Pólya. *How to Solve It: A New Aspect of Mathematical Method*. Princeton University Press, Princeton, NJ, 1945. doi: 10.1515/9781400828678.

[86] K. R. Popper. *The Logic of Scientific Discovery*. Hutchinson, London, 1959. doi: 10.4324/9780203994627.

[87] K. R. Popper. *Conjectures and Refutations: The Growth of Scientific Knowledge*. Routledge & Kegan Paul, London, 1963. doi: 10.4324/9780203538074.

[88] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763, 2021. doi: 10.48550/arXiv.2103.00020. URL https://proceedings.mlr.press/v139/radford21a.html.

[89] E. Real, C. Liang, D. R. So, and Q. V. Le. AutoML-Zero: Evolving machine learning algorithms from scratch. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, volume 119 of *Proceedings of Machine Learning Research*, pages 8007–8019, 2020. doi: 10.48550/arXiv.2003.03384. URL https://proceedings.mlr.press/v119/real20a.html.

[90] J. R. Rice. The algorithm selection problem. In M. Rubinoff and M. C. Yovits, editors, *Advances in Computers*, volume 15, pages 65–118. Academic Press, New York, 1976. doi: 10.1016/S0065-2458(08)60520-3.

[91] B. Romera-Paredes, M. Barekatain, A. Novikov, M. Balog, M. P. Kumar, E. Dupont, F. J. R. Ruiz, J. S. Ellenberg, P. Wang, O. Fawzi, P. Kohli, and A. Fawzi. Mathematical discoveries from program search with large language models. *Nature*, 625(7995):468–475, 2024. doi: 10.1038/s41586-023-06924-6.

[92] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, A. C. Berg, and L. Fei-Fei. ImageNet large scale visual recognition challenge. *International Journal of Computer Vision*, 115(3):211–252, 2015. doi: 10.1007/s11263-015-0816-y.

[93] A. L. Samuel. Some studies in machine learning using the game of checkers. *IBM Journal of Research and Development*, 3(3):210–229, 1959. doi: 10.1147/rd.33.0210.

[94] J. Schmidhuber. Gödel machines: Self-referential universal problem solvers making provably optimal self-improvements. arXiv:cs/0309048, 2003. URL https://arxiv.org/abs/cs/0309048.

[95] J. Schrittwieser, I. Antonoglou, T. Hubert, K. Simonyan, L. Sifre, S. Schmitt, A. Guez, E. Lockhart, D. Hassabis, T. Graepel, T. Lillicrap, and D. Silver. Mastering Atari, Go, chess and shogi by planning with a learned model. *Nature*, 588(7839):604–609, 2020. doi: 10.1038/s41586-020-03051-4.

[96] C. E. Shannon. Programming a computer for playing chess. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 41(314):256–275, 1950. doi: 10.1080/14786445008521796.

[97] M. Shin, J. Kim, B. van Opheusden, and T. L. Griffiths. Superhuman artificial intelligence can improve human decision-making by increasing novelty. *Proceedings of the National Academy of Sciences*, 120(12):e2214840120, 2023. doi: 10.1073/pnas.2214840120.

[98] C. Si, D. Yang, and T. Hashimoto. Can LLMs generate novel research ideas? A large-scale human study with 100+ NLP researchers. In *International Conference on Learning Representations (ICLR)*, 2025. doi: 10.48550/arXiv.2409.04109. URL https://openreview.net/forum?id=M23dTGWCZy.

[99] C. Si, T. Hashimoto, and D. Yang. The ideation–execution gap: Execution outcomes of LLM-generated versus human research ideas. In *International Conference on Learning Representations (ICLR)*, 2026. doi: 10.48550/arXiv.2506.20803. URL https://openreview.net/forum?id=Fllp8l6Puy.

[100] D. Silver, A. Huang, C. J. Maddison, A. Guez, L. Sifre, G. van den Driessche, J. Schrittwieser, I. Antonoglou, V. Panneershelvam, M. Lanctot, S. Dieleman, D. Grewe, J. Nham, N. Kalchbrenner, I. Sutskever, T. Lillicrap, M. Leach, K. Kavukcuoglu, T. Graepel, and D. Hassabis. Mastering the game of Go with deep neural networks and tree search. *Nature*, 529(7587):484–489, 2016. doi: 10.1038/nature16961.

[101] D. Silver, J. Schrittwieser, K. Simonyan, I. Antonoglou, A. Huang, A. Guez, T. Hubert, L. Baker, M. Lai, A. Bolton, Y. Chen, T. Lillicrap, F. Hui, L. Sifre, G. van den Driessche, T. Graepel, and D. Hassabis. Mastering the game of Go without human knowledge. *Nature*, 550 (7676):354–359, 2017. doi: 10.1038/nature24270.

[102] D. Silver, T. Hubert, J. Schrittwieser, I. Antonoglou, M. Lai, A. Guez, M. Lanctot, L. Sifre, D. Kumaran, T. Graepel, T. Lillicrap, K. Simonyan, and D. Hassabis. A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play. *Science*, 362(6419):1140–1144, 2018. doi: 10.1126/science.aar6404. URL https://arxiv.org/abs/1712.01815. Earlier version: arXiv:1712.01815 (2017). Training statistics: supplementary Table S3.

[103] D. K. Simonton. Creativity and discovery as blind variation: Campbell's (1960) BVSR model after the half-century mark. *Review of General Psychology*, 15(2):158–174, 2011. doi: 10.1037/a0022912.

[104] R. Socher. AI's biggest breakthrough won't be a chatbot. Next Big Idea Club, Book Bites, 8 October 2026. URL https://nextbigideaclub.com/magazine/ais-biggest-breakthrough-wont-chatbot-bookbite/61534/.

[105] R. Socher. *The Eureka Machine: Why AI Is the Key to Unlocking a New Era of Scientific Discoveries*. PublicAffairs, New York, 2026. ISBN 9781541705708. URL https://www.hachettebookgroup.com/titles/richard-socher/the-eureka-machine/9781541705708/.

[106] J. M. Stokes, K. Yang, K. Swanson, W. Jin, A. Cubillos-Ruiz, N. M. Donghia, C. R. MacNair, S. French, L. A. Carfrae, Z. Bloom-Ackermann, V. M. Tran, A. Chiappino-Pepe, A. H. Badran, I. W. Andrews, E. J. Chory, G. M. Church, E. D. Brown, T. S. Jaakkola, R. Barzilay, and J. J. Collins. A deep learning approach to antibiotic discovery. *Cell*, 180(4):688–702.e13, 2020. doi: 10.1016/j.cell.2020.01.021.

[107] A. Surina, A. Mansouri, L. Quaedvlieg, A. Seddas, M. Viazovska, E. Abbe, and C. Gulcehre. Algorithm discovery with LLMs: Evolutionary search meets reinforcement learning. arXiv:2504.05108, 2025. URL https://arxiv.org/abs/2504.05108.

[108] R. S. Sutton. The bitter lesson. Essay, 13 March 2019. URL http://www.incompleteideas.net/IncIdeas/BitterLesson.html.

[109] G. Tesauro. Temporal difference learning and TD-Gammon. *Communications of the ACM*, 38(3):58–68, 1995. doi: 10.1145/203330.203343.

[110] T. H. Trinh, Y. Wu, Q. V. Le, H. He, and T. Luong. Solving olympiad geometry without human demonstrations. *Nature*, 625(7995): 476–482, 2024. doi: 10.1038/s41586-023-06747-5.

[111] A. van den Oord, Y. Li, and O. Vinyals. Representation learning with contrastive predictive coding. arXiv:1807.03748, 2018. URL https://arxiv.org/abs/1807.03748.

[112] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, pages 5998–6008, 2017. doi: 10.48550/arXiv.1706.03762. URL https://arxiv.org/abs/1706.03762.

[113] O. Vinyals, I. Babuschkin, W. M. Czarnecki, M. Mathieu, A. Dudzik, J. Chung, D. H. Choi, R. Powell, T. Ewalds, P. Georgiev, J. Oh, D. Horgan, M. Kroiss, I. Danihelka, A. Huang, L. Sifre, T. Cai, J. P. Agapiou, M. Jaderberg, A. S. Vezhnevets, R. Leblond, T. Pohlen, V. Dalibard, D. Budden, Y. Sulsky, J. Molloy, T. L. Paine, C. Gulcehre,

Z. Wang, T. Pfaff, Y. Wu, R. Ring, D. Yogatama, D. Wünsch, K. McKinney, O. Smith, T. Schaul, T. Lillicrap, K. Kavukcuoglu, D. Hassabis, C. Apps, and D. Silver. Grandmaster level in StarCraft II using multi-agent reinforcement learning. *Nature*, 575(7782): 350–354, 2019. doi: 10.1038/s41586-019-1724-z.

[114] S. Von Arx, L. Chan, and B. Barnes. Recent frontier models are reward hacking. METR blog, 5 June 2025. URL https://metr.org/blog/2025-06-05-recent-reward-hacking/.

[115] G. Wallas. *The Art of Thought*. Jonathan Cape, London, 1926.

[116] K. Wiggers. Sakana walks back claims that its AI can dramatically speed up model training. TechCrunch, 21 February 2025. URL https://techcrunch.com/2025/02/21/sakana-walks-back-claims-that-its-ai-can-dramatically-speed-up-model-training/.

[117] T. Wolf. The Einstein AI model. Technical blog essay, February 2025. URL https://thomwolf.io/blog/scientific-ai.html.

[118] D. H. Wolpert. The lack of a priori distinctions between learning algorithms. *Neural Computation*, 8(7):1341–1390, 1996. doi: 10.1162/neco.1996.8.7.1341.

[119] M. Yuksekgonul, D. Koceja, X. Li, F. Bianchi, J. McCaleb, X. Wang, J. Kautz, Y. Choi, J. Zou, C. Guestrin, and Y. Sun. Learning to discover at test time. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026. doi: 10.48550/arXiv.2601.16175. URL https://arxiv.org/abs/2601.16175.

[120] E. Zelikman, E. Lorch, L. Mackey, and A. T. Kalai. Self-Taught Optimizer (STOP): Recursively self-improving code generation. In *Conference on Language Modeling (COLM)*, 2024. doi: 10.48550/arXiv.2310.02304. URL https://arxiv.org/abs/2310.02304.

[121] J. Zhang, S. Hu, C. Lu, R. Lange, and J. Clune. Darwin Gödel Machine: Open-ended evolution of self-improving agents. arXiv:2505.22954, 2025. URL https://arxiv.org/abs/2505.22954.

[122] A. Zhao, Y. Wu, Y. Yue, T. Wu, Q. Xu, M. Lin, S. Wang, Q. Wu, Z. Zheng, and G. Huang. Absolute Zero: Reinforced self-play reasoning with zero data. arXiv:2505.03335, 2025. URL https://arxiv.org/abs/2505.03335.

[123] Z. Zhong, I. Segal, I. Bercovich, S. Saxena, K. Zhang, and A. Raghunathan. Hardening agent benchmarks with adversarial hacker-fixer loops. arXiv:2606.08960, 2026. URL https://arxiv.org/abs/2606.08960.

[124] B. Zoph and Q. V. Le. Neural architecture search with reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2017. doi: 10.48550/arXiv.1611.01578. URL https://arxiv.org/abs/1611.01578.